

Planning and Self-Reflection in Large Language Model Reasoning: A Survey

Planning and self-reflection have become central strategies for improving LLM reasoning beyond single-pass generation. Planning decomposes goals, explores alternatives, and allocates computation across intermediate steps; self-reflection critiques partial solutions, repairs errors, and uses feedback from verifiers, tools, or environments. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around chain-of-thought decomposition, search over reasoning traces, tree and graph deliberation, self-correction, verifier-guided revision, agent planning, training, benchmarks, and robustness. The survey argues that planning and reflection are complementary control mechanisms: planning structures the future trajectory, while reflection updates beliefs about past steps. Their promise is greater reliability and sample efficiency; their risks are unfaithful rationales, overthinking, reward hacking, and evaluation artifacts. We close with prospects for calibrated, auditable, and environment-grounded reasoning.

Keywords:

large language models; planning; self-reflection; reasoning; chain-of-thought; verifiers; agents

1 Introduction

Large language models often solve problems by producing a single response, but many tasks require multi-step reasoning, search, revision, and feedback. Planning and self-reflection are two families of methods that expose this intermediate control structure. Planning asks the model to decompose goals, propose substeps, allocate tools or computations, and choose a trajectory. Self-reflection asks the model or an auxiliary evaluator to inspect intermediate outputs, identify errors, and revise the next attempt [8, 21, 65, 82, 118, 145, 159, 220]. Together, they convert reasoning from a one-shot generation problem into an iterative decision process.

The distinction is useful but not absolute. A plan may include reflection checkpoints; a reflection step may generate a new plan; and agentic systems often interleave reasoning, action, observation, and critique. Chain-of-thought prompting made intermediate reasoning visible, while tree, graph, and search-based methods made alternatives explicit [9, 91, 106, 127, 161, 234, 241, 292]. Self-correction and verifier-guided methods then added a feedback chan-

nel that can reject, rank, or repair candidate traces [8, 58, 65, 82, 145, 220, 240, 241]. The modern literature explores how to combine these mechanisms without producing verbose but unfaithful rationales.

This survey treats planning and self-reflection as test-time control mechanisms for LLM reasoning. They are not merely prompting tricks; they define how computation is allocated, how uncertainty is handled, how errors are detected, and how the model interacts with external feedback. The central question is when extra reasoning improves correctness, and when it amplifies bias, spurious rationalization, or cost.

2 Background

2.1 Reasoning Traces and Their Ambiguity

Intermediate reasoning traces can help models solve harder tasks and help humans inspect outputs. However, a trace is not guaranteed to be faithful to the computation that produced the answer. LLMs can generate plausible explanations after the fact, and

longer traces can hide errors in a stream of confident language. Planning and reflection methods must therefore be evaluated by outcomes, process quality, and trace faithfulness rather than by fluency alone [19, 30, 39, 65, 91, 159, 241, 244].

2.2 Planning as Test-Time Search

Planning expands the model’s computation beyond a single greedy answer. It can decompose a task, sample candidate steps, search over partial solutions, call tools, or evaluate branches. Search-based reasoning is useful when local decisions are uncertain and when a verifier can score partial or final states. Its costs are latency, token budget, and the possibility that search explores many superficially plausible but invalid traces [19, 39, 65, 118, 193, 226, 234, 240].

2.3 Reflection as Feedback

Reflection introduces a feedback loop. The model may critique its own answer, compare alternatives, inspect tool output, or revise after a verifier flags an error. Reflection is effective when the feedback signal contains information not already used in the first attempt. It is less effective when the model simply repeats its assumptions or rationalizes a wrong answer. This makes verifier design, environment feedback, and uncertainty calibration central to reflective reasoning [8, 21, 30, 123, 145, 161, 234, 292].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the topic with planning, self-reflection, chain-of-thought, tree-of-thought, graph-of-thought, self-correction, verifiers, process supervision, ReAct, Reflexion, mathematical reasoning, code reasoning, planning benchmarks, reward models, test-time compute, and deliberation terms. Records were retained when title or abstract evidence connected LLMs or language models with planning, reasoning traces, reflection, critique, verification, agent feedback, training, evaluation, or robustness. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize methods along five axes. The first is *trace structure*: linear chain, tree, graph, program,

plan, or trajectory. The second is *control policy*: prompting, sampling, search, learned planner, external solver, or agent loop. The third is *feedback source*: self-critique, another model, symbolic verifier, unit test, tool observation, human preference, or environment reward. The fourth is *revision target*: answer, reasoning trace, plan, tool choice, memory, or model parameters. The fifth is *evaluation*: final accuracy, process correctness, cost, latency, calibration, and robustness.

3.3 Decomposition and Chain-of-Thought

Decomposition methods make intermediate reasoning explicit. Chain-of-thought prompting asks the model to produce a sequence of steps; least-to-most and related strategies break complex tasks into subproblems; scratchpads externalize state. These methods are simple and broadly applicable, but they rely on the model to maintain local validity across steps. They work best when the task naturally decomposes and when errors can be detected before they contaminate later steps [9, 91, 106, 127, 161, 234, 241, 292].

3.4 Tree, Graph, and Search-Based Reasoning

Search-based methods sample multiple candidate thoughts, branch over alternatives, and use scores to select promising paths. Tree and graph formulations make dependencies among intermediate states explicit. They can improve hard reasoning tasks by allocating more test-time computation, but the benefit depends on proposal diversity and evaluator quality. Poor evaluators can make search confidently select bad traces, while excessive branching can make inference impractically expensive [19, 39, 65, 118, 193, 226, 234, 240].

3.5 Self-Reflection and Self-Correction

Self-reflection methods ask the model to examine a prior attempt and produce a critique, diagnosis, or revised answer. Some approaches run multiple critique-revise cycles; others store reflections as memory for future tasks. Reflection is most useful when it changes the information state: for example, after a failed unit test, a contradictory observation, or a comparison with retrieved evidence. Reflection without new evidence can still help by encouraging error search, but it also risks reinforcing mistaken premises [8, 58, 65, 82, 145, 220, 240, 241].

3.6 Verifiers, Reward Models, and Process Supervision

Verifiers provide an external or learned signal for judging candidate reasoning. They may check final answers, score intermediate steps, execute code, validate equations, or rank traces. Outcome supervision is easier to collect but may miss invalid intermediate reasoning; process supervision can guide search more directly but is expensive and domain specific. Reward models and judges extend this idea to open-ended reasoning, where correctness is harder to formalize [8, 21, 30, 123, 145, 161, 234, 292].

3.7 Agent Planning and Environment Feedback

In agent settings, planning and reflection are grounded by actions and observations. The model proposes a plan, calls tools or interacts with an environment, observes results, and revises. This setting exposes reasoning to external feedback, which can make reflection more meaningful than purely textual self-critique. It also introduces new errors: invalid tool calls, stale memory, irreversible actions, and trajectories that achieve a local subgoal while violating the user’s broader intent [21, 58, 65, 82, 118, 145, 159, 220].

3.8 Training and Adaptation

Planning and reflection can be elicited at test time, but training can improve when and how they are used. Supervised fine-tuning teaches trace formats; reinforcement learning can reward correct answers or efficient plans; preference optimization can favor helpful critiques; and distillation can compress expensive deliberation into smaller models. The open problem is to train models that deliberate when useful and answer directly when additional reasoning is unnecessary [58, 82, 109, 145, 159, 193, 240, 292].

3.9 Benchmarks and Evaluation

Evaluation must separate final-answer accuracy from reasoning quality. Mathematical, logical, code, planning, and agentic benchmarks stress different aspects of planning and reflection. Good evaluations track cost, number of attempts, verifier calls, trace validity, and robustness to misleading intermediate feedback. Benchmarks should also detect overthinking: cases where extra reflection reduces accuracy or increases unsupported rationalization [21, 58, 82, 118, 145, 159, 220, 240].

3.10 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

Decomposition and chain-of-thought reasoning The decomposition and chain-of-thought reasoning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [40, 76, 84, 91, 96, 161, 179, 190, 299]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [52, 77, 92, 94, 130, 147, 225, 249, 263]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [26, 32, 66, 67, 182, 191, 231, 267, 282]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [50, 62, 63, 75, 140, 143, 208, 257, 276]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [160, 256].

Search, tree, and graph reasoning The search, tree, and graph reasoning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [49, 60, 137, 149, 193, 209, 248, 265, 294]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [3, 4, 22, 80, 107, 148, 255, 277, 287]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [69, 158, 168, 258].

Self-reflection, self-correction, and critique The self-reflection, self-correction, and critique cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [8, 19, 30, 47, 123, 127, 145, 259, 274]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [95, 97, 155, 183, 200, 204, 210, 247, 285]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [24, 28, 41, 44, 51, 202, 207, 217, 271]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [11, 18, 59, 101, 126, 132, 175, 186, 187]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions

[7, 43, 116, 129, 133, 154, 170, 238, 297]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [100, 178, 181, 184, 192, 232, 237, 251, 279]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [1, 31, 114, 119, 169, 176, 194, 286, 300]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [13, 71, 72, 99, 113, 180, 185, 266, 298]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [10, 12, 120, 171, 199, 235, 272, 275, 284]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [157].

Verifiers, evaluators, and reward models The verifiers, evaluators, and reward models cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [93, 213].

Agent planning and environment feedback The agent planning and environment feedback cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [39, 65, 82, 118, 226, 234, 240, 241, 292]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [9, 139, 141, 150, 214, 244, 245, 281, 289]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [2, 14, 74, 110, 163, 198, 212, 218, 293]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [111, 125, 146, 173, 195, 236, 260, 291, 296]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [5, 37, 48, 89, 90, 138, 142, 165, 227]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [35, 38, 103, 136, 164, 197, 203, 211].

Training and reinforcement learning for reasoning The training and reinforcement learning for reasoning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [57, 105, 131, 174, 221, 228, 252–254]. A second strand in this cluster studies how

such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [27, 45, 78, 122, 216].

Benchmarks and domains for reasoning The benchmarks and domains for reasoning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [21, 58, 85, 106, 109, 112, 159, 220, 229]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [20, 73, 86, 88, 189, 230, 239, 250, 278]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [16, 25, 34, 104, 124, 166, 167, 280, 290]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [79, 87, 98, 156, 172, 224, 242, 246, 264]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [15, 36, 61, 102, 135, 152, 153, 188, 201]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [6, 33, 53, 68, 83, 215, 219, 262, 270]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [56, 70, 108, 117, 128, 177, 223, 288, 295]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [17, 29, 81, 134, 206, 233, 261, 273, 283]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [42, 54, 64, 115, 121, 162, 196, 222, 269]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [23, 46, 55, 144, 151, 205, 243, 268].

4 Challenges and Prospects

4.1 Faithfulness of Reasoning Traces

A central challenge is determining whether generated reasoning explains the answer or merely accompanies it. Future work needs interventions, counterfactual tests, verifier audits, and process-level datasets that can distinguish faithful plans from persuasive narratives.

4.2 When to Deliberate

Planning and reflection consume tokens, time, and sometimes tool calls. The best system is not the one that always thinks longer, but the one that allocates deliberation adaptively. Confidence estimates, task difficulty predictors, and cheap verifiers may help decide when extra reasoning is warranted.

4.3 Verifier Reliability

Search and reflection inherit the biases of their evaluators. A flawed verifier can reward superficial traces, penalize creative valid solutions, or encourage reward hacking. Robust reasoning systems need verifier ensembles, calibrated uncertainty, and fallback mechanisms when evaluators disagree [8, 21, 30, 123, 145, 161, 234, 292].

4.4 Grounded Reflection

Reflection is more reliable when grounded in external evidence: tests, tools, simulations, retrieved documents, or human feedback. The field is likely to move from purely textual self-critique toward environment-grounded reflection, where revision is tied to observable failure modes.

5 Conclusion

Planning and self-reflection extend LLM reasoning by making intermediate computation explicit and revisable. This survey reviewed decomposition, chain-of-thought, tree and graph search, self-correction, verifiers, agent planning, training, benchmarks, and robustness. The central lesson is that extra reasoning is valuable only when guided by reliable feedback and calibrated control. Future systems should deliberate adaptively, expose auditable traces, and ground reflection in evidence rather than verbosity.

References

- [1] Sara Abdali, Can Goksen, Michael Solodko, Saeed Amizadeh, Julie E. Maybee, and Kazuhito Koishida. Self-reflecting Large Language Models: A Hegelian Dialectical Approach. *arXiv preprint arXiv:2501.14917*, 2025. URL <https://arxiv.org/abs/2501.14917>.
- [2] Krish Agarwal, Yuqian Jiang, Jiaheng Hu, Bo Liu, and Peter Stone. L3M+P: Lifelong Planning with Large Language Models. *arXiv preprint arXiv:2508.01917*, 2025. URL <https://arxiv.org/abs/2508.01917>.
- [3] Mohamed Aghzal, Erion Plaku, and Ziyu Yao. Can Large Language Models be Good Path Planners? A Benchmark and Investigation on Spatial-temporal Reasoning. *arXiv preprint arXiv:2310.03249*, 2023. URL <https://arxiv.org/abs/2310.03249>.
- [4] Mohamed Aghzal, Erion Plaku, Gregory J. Stein, and Ziyu Yao. A Survey on Large Language Models for Automated Planning. *arXiv preprint arXiv:2502.12435*, 2025. URL <https://arxiv.org/abs/2502.12435>.
- [5] Anurag Ajay, Seungwook Han, Yilun Du, Shuang Li, Abhi Gupta, Tommi Jaakkola, Josh Tenenbaum, Leslie Kaelbling, Akash Srivastava, and Pulkit Agrawal. Compositional Foundation Models for Hierarchical Planning. *arXiv preprint arXiv:2309.08587*, 2023. URL <https://arxiv.org/abs/2309.08587>.
- [6] Md Fahim Anjum. When Reasoning Beats Scale: A 1.5B Reasoning Model Outranks 13B LLMs as Discriminator. *arXiv preprint arXiv:2505.03786*, 2025. URL <https://arxiv.org/abs/2505.03786>.
- [7] Riku Arakawa and Hiromu Yakura. Coaching Copilot: Blended Form of an LLM-Powered Chatbot and a Human Coach to Effectively Support Self-Reflection for Leadership Growth. *arXiv preprint arXiv:2405.15250*, 2024. URL <https://arxiv.org/abs/2405.15250>.
- [8] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *arXiv preprint arXiv:2310.11511*, 2023. URL <https://arxiv.org/abs/2310.11511>.
- [9] David Bai, Ishika Singh, David Traum, and Jesse Thomason. TwoStep: Multi-agent Task Planning using Classical Planners and Large Language Models. *arXiv preprint arXiv:2403.17246*, 2024. URL <https://arxiv.org/abs/2403.17246>.
- [10] Yuzhuo Bai, Shitong Duan, Muhua Huang, Jing Yao, Zhenghao Liu, Peng Zhang, Tun Lu, Xiaoyuan Yi, Maosong Sun, and Xing Xie. IROTE: Human-like Traits Elicitation of Large Language Model via In-Context Self-Reflective Optimization. *arXiv preprint arXiv:2508.08719*, 2025. URL <https://arxiv.org/abs/2508.08719>.
- [11] Adrian Bazaga, Rexhina Blloshmi, Bill Byrne, and Adria de Gispert. Learning to Reason Over Time: Timeline Self-Reflection for Improved Temporal Reasoning in Language Models. *arXiv preprint arXiv:2504.05258*, 2025. URL <https://arxiv.org/abs/2504.05258>.
- [12] Cameron Berg, Diogo de Lucena, and Judd Rosenblatt. Large Language Models Report Subjective Experience Under Self-Referential Processing.

- arXiv preprint arXiv:2510.24797*, 2025. URL <https://arxiv.org/abs/2510.24797>.
- [13] Leonardo Berti, Flavio Giorgi, and Gjergji Kasneci. Emergent Abilities in Large Language Models: A Survey. *arXiv preprint arXiv:2503.05788*, 2025. URL <https://arxiv.org/abs/2503.05788>.
 - [14] Timo Birr, Christoph Pohl, Abdelrahman Younes, and Tamim Asfour. AutoGPT+P: Affordance-based Task Planning with Large Language Models. *arXiv preprint arXiv:2402.10778*, 2024. URL <https://arxiv.org/abs/2402.10778>.
 - [15] Bernd Bohnet, Azade Nova, Aaron T Parisi, Kevin Swersky, Katayoon Goshvadi, Hanjun Dai, Dale Schuurmans, Noah Fiedel, and Hanie Sedghi. Exploring and Benchmarking the Planning Capabilities of Large Language Models. *arXiv preprint arXiv:2406.13094*, 2024. URL <https://arxiv.org/abs/2406.13094>.
 - [16] Bernd Bohnet, Michael C. Mozer, Kevin Swersky, Wil Cunningham, Aaron Parisi, Kathleen Keane, and Noah Fiedel. Analysis of Optimality of Large Language Models on Planning Problems. *arXiv preprint arXiv:2604.02910*, 2026. URL <https://arxiv.org/abs/2604.02910>.
 - [17] Abdessalam Boucekif, Samer Rashwani, Heba Sbahi, Shahd Gaben, Mutaz Al-Khatib, and Mohammed Ghaly. Assessing Large Language Models on Islamic Legal Reasoning: Evidence from Inheritance Law Evaluation. *arXiv preprint arXiv:2509.01081*, 2025. URL <https://arxiv.org/abs/2509.01081>.
 - [18] Kecheng Cai, Chenyang Xu, Chao Peng, Jiafu Huang, Qiyuan Liang, and Irene Zheng. Combating Spurious Correlations in Graph Interpretability via Self-Reflection. *arXiv preprint arXiv:2601.11021*, 2026. URL <https://arxiv.org/abs/2601.11021>.
 - [19] Yishuai Cai, Xinglin Chen, Yunxin Mao, Minglong Li, Shaowu Yang, Wenjing Yang, and Ji Wang. HBTP: Heuristic Behavior Tree Planning with Large Language Model Reasoning. *arXiv preprint arXiv:2406.00965*, 2024. URL <https://arxiv.org/abs/2406.00965>.
 - [20] Pengfei Cao, Tianyi Men, Wencan Liu, Jingwen Zhang, Xuzhao Li, Xixun Lin, Dianbo Sui, Yanan Cao, Kang Liu, and Jun Zhao. Large Language Models for Planning: A Comprehensive and Systematic Survey. *arXiv preprint arXiv:2505.19683*, 2025. URL <https://arxiv.org/abs/2505.19683>.
 - [21] Zouying Cao, Runze Wang, Yifei Yang, Xinbei Ma, Xiaoyong Zhu, Bo Zheng, and Hai Zhao. PGPO: Enhancing Agent Reasoning via Pseudocode-style Planning Guided Preference Optimization. *arXiv preprint arXiv:2506.01475*, 2025. URL <https://arxiv.org/abs/2506.01475>.
 - [22] Alessio Capitanelli and Fulvio Mastrogiovanni. A Framework for Neurosymbolic Robot Action Planning using Large Language Models. *arXiv preprint arXiv:2303.00438*, 2023. URL <https://arxiv.org/abs/2303.00438>.
 - [23] Fu-Chieh Chang, You-Chen Lin, and Pei-Yuan Wu. Unraveling Arithmetic in Large Language Models: The Role of Algebraic Structures. *arXiv preprint arXiv:2411.16260*, 2024. URL <https://arxiv.org/abs/2411.16260>.
 - [24] Guoxin Chen, Zhong Zhang, Xin Cong, Fangda Guo, Yesai Wu, Yankai Lin, Wenzheng Feng, and Yasheng Wang. Learning Evolving Tools for Large Language Models. *arXiv preprint arXiv:2410.06617*, 2024. URL <https://arxiv.org/abs/2410.06617>.
 - [25] Jianghao Chen, Zhenlin Wei, Zhenjiang Ren, Ziyong Li, and Jiajun Zhang. LR²Bench: Evaluating Long-chain Reflective Reasoning Capabilities of Large Language Models via Constraint Satisfaction Problems. *arXiv preprint arXiv:2502.17848*, 2025. URL <https://arxiv.org/abs/2502.17848>.
 - [26] Jingyun Chen, Linghan Cai, Zhikang Wang, Yi Huang, Songhan Jiang, Shenjin Huang, Hongpeng Wang, and Yongbing Zhang. PathAgent: Toward Interpretable Analysis of Whole-slide Pathology Images via Large Language Model-based Agentic Reasoning. *arXiv preprint arXiv:2511.17052*, 2025. URL <https://arxiv.org/abs/2511.17052>.
 - [27] Kedi Chen, Zhikai Lei, Xu Guo, Xuecheng Wu, Siyuan Zeng, Jianghao Yin, Yinqi Zhang, Qin Chen, Jie Zhou, Liang He, Qipeng Guo, Kai Chen, and Wei Zhang. Code-driven Number Sequence Calculation: Enhancing the inductive Reasoning Abilities of Large Language Models. *arXiv preprint arXiv:2510.14620*, 2025. URL <https://arxiv.org/abs/2510.14620>.
 - [28] Liyi Chen, Panrong Tong, Zhongming Jin, Ying Sun, Jieping Ye, and Hui Xiong. Plan-on-Graph: Self-Correcting Adaptive Planning of Large Language Model on Knowledge Graphs. *arXiv preprint arXiv:2410.23875*, 2024. URL <https://arxiv.org/abs/2410.23875>.
 - [29] Meilin Chen, Zepeng Zhai, Jiaxuan Zhao, and Yuan Lu. SafeRun: Enabling Determinism in LLM Planning for Running. *arXiv preprint arXiv:2606.09027*, 2026. URL <https://arxiv.org/abs/2606.09027>.
 - [30] Meiqi Chen, Fandong Meng, Yingxue Zhang, Yan Zhang, and Jie Zhou. CRAT: A Multi-Agent Framework for Causality-Enhanced Reflective and Retrieval-Augmented Translation with Large Language Models. *arXiv preprint arXiv:2410.21067*, 2024. URL <https://arxiv.org/abs/2410.21067>.

- [31] Qiguang Chen, Hanjing Li, Libo Qin, Dengyun Peng, Jinhao Liu, Jiangyi Wang, Chengyue Wu, Xie Chen, Yantao Du, and Wanxiang Che. Beyond Surface Reasoning: Unveiling the True Long Chain-of-Thought Capacity of Diffusion Large Language Models. *arXiv preprint arXiv:2510.09544*, 2025. URL <https://arxiv.org/abs/2510.09544>.
- [32] Qiguang Chen, Libo Qin, Jinhao Liu, Dengyun Peng, Jiannan Guan, Peng Wang, Mengkang Hu, Yuhang Zhou, Te Gao, and Wanxiang Che. Towards Reasoning Era: A Survey of Long Chain-of-Thought for Reasoning Large Language Models. *arXiv preprint arXiv:2503.09567*, 2025. URL <https://arxiv.org/abs/2503.09567>.
- [33] Runjin Chen, Zhenyu Zhang, Junyuan Hong, Souvik Kundu, and Zhangyang Wang. SEAL: Steerable Reasoning Calibration of Large Language Models for Free. *arXiv preprint arXiv:2504.07986*, 2025. URL <https://arxiv.org/abs/2504.07986>.
- [34] Sijia Chen and Di Niu. iCLP: Large Language Model Reasoning with Implicit Cognition Latent Planning. *arXiv preprint arXiv:2512.24014*, 2025. URL <https://arxiv.org/abs/2512.24014>.
- [35] Siwei Chen, Anxing Xiao, and David Hsu. LLM-State: Open World State Representation for Long-horizon Task Planning with Large Language Model. *arXiv preprint arXiv:2311.17406*, 2023. URL <https://arxiv.org/abs/2311.17406>.
- [36] Yi Chen, Yuying Ge, Yixiao Ge, Mingyu Ding, Bohao Li, Rui Wang, Ruifeng Xu, Ying Shan, and Xihui Liu. EgoPlan-Bench: Benchmarking Multimodal Large Language Models for Human-Level Planning. *arXiv preprint arXiv:2312.06722*, 2023. URL <https://arxiv.org/abs/2312.06722>.
- [37] Yibin Chen, Yifu Yuan, Zeyu Zhang, Yan Zheng, Jinyi Liu, Fei Ni, Jianye Hao, Hangyu Mao, and Fuzheng Zhang. SheetAgent: Towards A Generalist Agent for Spreadsheet Reasoning and Manipulation via Large Language Models. *arXiv preprint arXiv:2403.03636*, 2024. URL <https://arxiv.org/abs/2403.03636>.
- [38] Yihan Chen, Benfeng Xu, Xiaorui Wang, Yongdong Zhang, and Zhendong Mao. Training LLM-Based Agents with Synthetic Self-Reflected Trajectories and Partial Masking. *arXiv preprint arXiv:2505.20023*, 2025. URL <https://arxiv.org/abs/2505.20023>.
- [39] Yiye Chen, Harpreet Sawhney, Nicholas Gyde, Yanan Jian, Jack Saunders, Patricio Vela, and Ben Lundell. Schema-Guided Scene-Graph Reasoning based on Multi-Agent Large Language Model System. *arXiv preprint arXiv:2502.03450*, 2025. URL <https://arxiv.org/abs/2502.03450>.
- [40] Kanzhi Cheng, Yantao Li, Fangzhi Xu, Jianbing Zhang, Hao Zhou, and Yang Liu. Vision-Language Models Can Self-Improve Reasoning via Reflection. *arXiv preprint arXiv:2411.00855*, 2024. URL <https://arxiv.org/abs/2411.00855>.
- [41] Nicholas Cheng. Reflective Translation: Improving Low-Resource Machine Translation via Structured Self-Reflection. *arXiv preprint arXiv:2601.19871*, 2026. URL <https://arxiv.org/abs/2601.19871>.
- [42] Hung-Ying Chu, Guan-Wei Chen, Shao-Yu Wei, and Yu-Cheng Lin. Large Language Models for 3D IC Space Planning. *arXiv preprint arXiv:2509.22716*, 2025. URL <https://arxiv.org/abs/2509.22716>.
- [43] Fan Cui, Chenyang Yin, Kexing Zhou, Youwei Xiao, Guangyu Sun, Qiang Xu, Qipeng Guo, Demin Song, Dahua Lin, Xingcheng Zhang, Yun, and Liang. OriGen: Enhancing RTL Code Generation with Code-to-Code Augmentation and Self-Reflection. *arXiv preprint arXiv:2407.16237*, 2024. URL <https://arxiv.org/abs/2407.16237>.
- [44] Fangming Cui, Ruixiao Zhu, Cheng Fang, Sunan Li, and Jiahong Li. Rethinking Agentic Reinforcement Learning In Large Language Models. *arXiv preprint arXiv:2604.27859*, 2026. URL <https://arxiv.org/abs/2604.27859>.
- [45] Zhenhe Cui, Huaxiang Xia, Hangjun Shen, Kailun Luo, Yong He, and Wei Liang. Abstraction Generation for Generalized Planning with Pre-trained Large Language Models. *arXiv preprint arXiv:2602.10485*, 2026. URL <https://arxiv.org/abs/2602.10485>.
- [46] Hanzheng Dai, Yuanliang Li, Jun Yan, and Zhibo Zhang. RefPentester: A Knowledge-Informed Self-Reflective Penetration Testing Framework Based on Large Language Models. *arXiv preprint arXiv:2505.07089*, 2025. URL <https://arxiv.org/abs/2505.07089>.
- [47] Andre de Souza Loureiro, Jorge Valverde-Rebaza, Julieta Noguez, David Escarcega, and Ricardo Marcacini. Advancing Multi-Step Mathematical Reasoning in Large Language Models through Multi-Layered Self-Reflection with Auto-Prompting. *arXiv preprint arXiv:2506.23888*, 2025. URL <https://arxiv.org/abs/2506.23888>.
- [48] Bin Deng, Yizhe Feng, Zeming Liu, Qing Wei, Xiangrong Zhu, Shuai Chen, Yuanfang Guo, and Yunhong Wang. RETAIL: Towards Real-world Travel Planning for Large Language Models. *arXiv preprint arXiv:2508.15335*, 2025. URL <https://arxiv.org/abs/2508.15335>.
- [49] Ziwei Deng, Mian Deng, Chenjing Liang, Zeming Gao, Chennan Ma, Chenxing Lin, Haipeng Zhang, Songzhu Mei, Siqi Shen, and Cheng Wang. PlanU: Large Language Model Reasoning through Planning under Uncertainty. *arXiv*

- preprint *arXiv:2510.18442*, 2025. URL <https://arxiv.org/abs/2510.18442>.
- [50] Hanna M. Dettki, Charley M. Wu, and Bob Rehder. Do LLMs Share Human-Like Biases? Causal Reasoning Under Prior Knowledge, Irrelevant Context, and Varying Compute Budgets. *arXiv preprint arXiv:2602.02983*, 2026. URL <https://arxiv.org/abs/2602.02983>.
 - [51] Rinku Dewri. GIER: Gap-Driven Self-Refinement for Large Language Models. *arXiv preprint arXiv:2509.00325*, 2025. URL <https://arxiv.org/abs/2509.00325>.
 - [52] Muhayy Ud Din, Waseem Akram, Ahsan B Bakht, Yihao Dong, and Irfan Hussain. Maritime Mission Planning for Unmanned Surface Vessel using Large Language Model. *arXiv preprint arXiv:2503.12065*, 2025. URL <https://arxiv.org/abs/2503.12065>.
 - [53] Jiayu Ding, Shuming Ma, Lei Cui, Nanning Zheng, and Furu Wei. LongReasonArena: A Long Reasoning Benchmark for Large Language Models. *arXiv preprint arXiv:2508.19363*, 2025. URL <https://arxiv.org/abs/2508.19363>.
 - [54] Zepeng Ding, Ruiyang Ke, Wenhao Huang, Guochao Jiang, Yanda Li, Deqing Yang, and Jiaqing Liang. Adaptive Reinforcement Learning Planning: Harnessing Large Language Models for Complex Information Extraction. *arXiv preprint arXiv:2406.11455*, 2024. URL <https://arxiv.org/abs/2406.11455>.
 - [55] Yubo Dong and Hehe Fan. Enhancing Large Language Models through Structured Reasoning. *arXiv preprint arXiv:2506.20241*, 2025. URL <https://arxiv.org/abs/2506.20241>.
 - [56] Zhichen Dong, Zhanhui Zhou, Zhixuan Liu, Chao Yang, and Chaochao Lu. Emergent Response Planning in LLMs. *arXiv preprint arXiv:2502.06258*, 2025. URL <https://arxiv.org/abs/2502.06258>.
 - [57] Zhihao Dou, Qinjian Zhao, Zhongwei Wan, Dinggen Zhang, Weida Wang, Towsif Raiyan, Benteng Chen, Qingtao Pan, Yang Ouyang, Chaoda Song, Zhiqiang Gao, Shufei Zhang, and Sumon Biswas. Plan Then Action: High-Level Planning Guidance Reinforcement Learning for LLM Reasoning. *arXiv preprint arXiv:2510.01833*, 2025. URL <https://arxiv.org/abs/2510.01833>.
 - [58] Yu Du, Fangyun Wei, and Hongyang Zhang. AnyTool: Self-Reflective, Hierarchical Agents for Large-Scale API Calls. *arXiv preprint arXiv:2402.04253*, 2024. URL <https://arxiv.org/abs/2402.04253>.
 - [59] Zackary Okun Dunivin, Mobina Noori, Seth Frey, and Curtis Atkinson. Self-reflection in Automated Qualitative Coding: Improving Text Annotation through Secondary LLM Critique. *arXiv preprint arXiv:2601.09905*, 2026. URL <https://arxiv.org/abs/2601.09905>.
 - [60] Hamza Ahmed Durrani and Rafay Suleman Durrani. Semantic Risk-Aware Heuristic Planning for Robotic Navigation in Dynamic Environments: An LLM-Inspired Approach. *arXiv preprint arXiv:2605.02862*, 2026. URL <https://arxiv.org/abs/2605.02862>.
 - [61] Abraham Paul Elenjical, Vivek Hruday Kavuri, and Vasudeva Varma. Think²: Grounded Metacognitive Reasoning in Large Language Models. *arXiv preprint arXiv:2602.18806*, 2026. URL <https://arxiv.org/abs/2602.18806>.
 - [62] Ye eun Cho. Evaluating Pragmatic Reasoning in Large Language Models: Evidence from Scalar Diversity. *arXiv preprint arXiv:2605.09042*, 2026. URL <https://arxiv.org/abs/2605.09042>.
 - [63] Bella Fascendini, Kathryn McGregor, Max D. Gupta, and Thomas L. Griffiths. The Riddle Riddle: Testing Flexible Reasoning in Large Language Models and Humans. *arXiv preprint arXiv:2606.27103*, 2026. URL <https://arxiv.org/abs/2606.27103>.
 - [64] Nick Ferguson, Liane Guillou, Alan Bundy, and Kwabena Nuamah. Evaluating the Meta- and Object-Level Reasoning of Large Language Models for Question Answering. *arXiv preprint arXiv:2502.10338*, 2025. URL <https://arxiv.org/abs/2502.10338>.
 - [65] Daocheng Fu, Jianbiao Mei, Licheng Wen, Xueming Yang, Cheng Yang, Rong Wu, Tao Hu, Siqi Li, Yufan Shen, Xinyu Cai, Pinlong Cai, Botian Shi, Yong Liu, and Yu Qiao. RE-Searcher: Robust Agentic Search with Goal-oriented Planning and Self-reflection. *arXiv preprint arXiv:2509.26048*, 2025. URL <https://arxiv.org/abs/2509.26048>.
 - [66] Fernando Gabriela Garcia, Spencer Burns, and Harrison Fuller. Leveraging Large Language Models for Comparative Literature Summarization with Reflective Incremental Mechanisms. *arXiv preprint arXiv:2412.02149*, 2024. URL <https://arxiv.org/abs/2412.02149>.
 - [67] Yuyao Ge, Shenghua Liu, Yiwei Wang, Lingrui Mei, Lizhe Chen, Baolong Bi, and Xueqi Cheng. Innate Reasoning is Not Enough: In-Context Learning Enhances Reasoning Large Language Models with Less Overthinking. *arXiv preprint arXiv:2503.19602*, 2025. URL <https://arxiv.org/abs/2503.19602>.
 - [68] Hejia Geng, Boxun Xu, and Peng Li. UPAR: A Kantian-Inspired Prompting Framework for Enhancing Large Language Model Capabilities. *arXiv preprint arXiv:2310.01441*, 2023. URL <https://arxiv.org/abs/2310.01441>.

- [69] Elliot Gestrin, Marco Kuhlmann, and Jendrik Seipp. NL2Plan: Robust LLM-Driven Planning from Minimal Text Descriptions. *arXiv preprint arXiv:2405.04215*, 2024. URL <https://arxiv.org/abs/2405.04215>.
- [70] Jonas Golde, Patrick Haller, Fabio Barth, and Alan Akbik. MastermindEval: A Simple But Scalable Reasoning Benchmark. *arXiv preprint arXiv:2503.05891*, 2025. URL <https://arxiv.org/abs/2503.05891>.
- [71] Junyi Gong, Zijie Qiu, and Ben Zhong Tang. Closing the Prior-Posterior Loop: Self-Reflective Molecular Design with Analysis-Driven LLM Iteration. *arXiv preprint arXiv:2606.09520*, 2026. URL <https://arxiv.org/abs/2606.09520>.
- [72] Hao Gu, Rui Zhong, Yu Xia, Wei Yang, Chi Lu, Peng Jiang, and Kun Gai. R4ec: A Reasoning, Reflection, and Refinement Framework for Recommendation Systems. *arXiv preprint arXiv:2507.17249*, 2025. URL <https://arxiv.org/abs/2507.17249>.
- [73] Jiayi Gui, Yiming Liu, Jiale Cheng, Xiaotao Gu, Xiao Liu, Hongning Wang, Yuxiao Dong, Jie Tang, and Minlie Huang. LogicGame: Benchmarking Rule-Based Reasoning Abilities of Large Language Models. *arXiv preprint arXiv:2408.15778*, 2024. URL <https://arxiv.org/abs/2408.15778>.
- [74] Huihui Guo, Huilong Pi, Yunchuan Qin, Zhuo Tang, and Kenli Li. Leveraging Pre-trained Large Language Models with Refined Prompting for Online Task and Motion Planning. *arXiv preprint arXiv:2504.21596*, 2025. URL <https://arxiv.org/abs/2504.21596>.
- [75] Yiduo Guo, Yaobo Liang, Chenfei Wu, Wenshan Wu, Dongyan Zhao, and Nan Duan. Learning to Plan with Natural Language. *arXiv preprint arXiv:2304.10464*, 2023. URL <https://arxiv.org/abs/2304.10464>.
- [76] Shibo Hao, Yi Gu, Haodi Ma, Joshua Jiahua Hong, Zhen Wang, Daisy Zhe Wang, and Zhiting Hu. Reasoning with Language Model is Planning with World Model. *arXiv preprint arXiv:2305.14992*, 2023. URL <https://arxiv.org/abs/2305.14992>.
- [77] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training Large Language Models to Reason in a Continuous Latent Space. *arXiv preprint arXiv:2412.06769*, 2024. URL <https://arxiv.org/abs/2412.06769>.
- [78] Teqi Hao, Xiaoyu Tan, Shaojie Shi, Yinghui Xu, and Xihe Qiu. Reflective Personalization Optimization: A Post-hoc Rewriting Framework for Black-Box Large Language Models. *arXiv preprint arXiv:2511.05286*, 2025. URL <https://arxiv.org/abs/2511.05286>.
- [79] Yilun Hao, Yongchao Chen, Yang Zhang, and Chuchu Fan. Large Language Models Can Solve Real-World Planning Rigorously with Formal Verification Tools. *arXiv preprint arXiv:2404.11891*, 2024. URL <https://arxiv.org/abs/2404.11891>.
- [80] Rishi Hazra, Pedro Zuidberg Dos Martires, and Luc De Raedt. SayCanPay: Heuristic Planning with Large Language Models using Learnable Domain Knowledge. *arXiv preprint arXiv:2308.12682*, 2023. URL <https://arxiv.org/abs/2308.12682>.
- [81] Chen He, Xun Jiang, Lei Wang, Hao Yang, Chong Peng, Peng Yan, Fumin Shen, and Xing Xu. What Makes Reasoning Invalid: Echo Reflection Mitigation for Large Language Models. *arXiv preprint arXiv:2511.06380*, 2025. URL <https://arxiv.org/abs/2511.06380>.
- [82] Chengbo He, Bochao Zou, Xin Li, Jiansheng Chen, Junliang Xing, and Huimin Ma. Enhancing LLM Reasoning with Multi-Path Collaborative Reactive and Reflection agents. *arXiv preprint arXiv:2501.00430*, 2024. URL <https://arxiv.org/abs/2501.00430>.
- [83] Haoyang He, Zihua Rong, Liangjie Zhao, Yunjia Zhao, Lan Yang, and Honggang Zhang. TTSR: Test-Time Self-Reflection for Continual Reasoning Improvement. *arXiv preprint arXiv:2603.03297*, 2026. URL <https://arxiv.org/abs/2603.03297>.
- [84] Jiashu He, Mingyu Derek Ma, Jinxuan Fan, Dan Roth, Wei Wang, and Alejandro Ribeiro. GIVE: Structured Reasoning of Large Language Models with Knowledge Graph Inspired Veracity Extrapolation. *arXiv preprint arXiv:2410.08475*, 2024. URL <https://arxiv.org/abs/2410.08475>.
- [85] Jie He, Victor Gutierrez-Basulto, and Jeff Z. Pan. From Sufficiency to Reflection: Reinforcement-Guided Thinking Quality in Retrieval-Augmented Reasoning for LLMs. *arXiv preprint arXiv:2507.22716*, 2025. URL <https://arxiv.org/abs/2507.22716>.
- [86] Qingrong He, Kejun Lin, Shizhe Chen, Anwen Hu, and Qin Jin. Think-Program-reCtify: 3D Situated Reasoning with Large Language Models. *arXiv preprint arXiv:2404.14705*, 2024. URL <https://arxiv.org/abs/2404.14705>.
- [87] Xusen Hei, Jiali Chen, Jinyu Yang, Mengchen Zhao, and Yi Cai. ViRectify: A Challenging Benchmark for Video Reasoning Correction with Multimodal Large Language Models. *arXiv preprint arXiv:2512.01424*, 2025. URL <https://arxiv.org/abs/2512.01424>.
- [88] Danny Hoang, David Gorsich, Matthew P. Castanier, and Farhad Imani. Knowledge Graph Fusion with Large Language Models for Accurate, Explainable Manufacturing Process Planning. *arXiv*

- preprint *arXiv:2506.13026*, 2025. URL <https://arxiv.org/abs/2506.13026>.
- [89] Jinbo Hou, Stefanos Bakirtzis, Kehai Qiu, Sichong Liao, Hui Song, Haonan Hu, Kezhi Wang, and Jie Zhang. iPLAN: Redefining Indoor Wireless Network Planning Through Large Language Models. *arXiv preprint arXiv:2507.19096*, 2025. URL <https://arxiv.org/abs/2507.19096>.
 - [90] Mengkang Hu, Yao Mu, Xinmiao Yu, Mingyu Ding, Shiguang Wu, Wenqi Shao, Qiguang Chen, Bin Wang, Yu Qiao, and Ping Luo. Tree-Planner: Efficient Close-loop Task Planning with Large Language Models. *arXiv preprint arXiv:2310.08582*, 2023. URL <https://arxiv.org/abs/2310.08582>.
 - [91] Minda Hu, Tianqing Fang, Jianshu Zhang, Junyu Ma, Zhisong Zhang, Jingyan Zhou, Hongming Zhang, Haitao Mi, Dong Yu, and Irwin King. WebCoT: Enhancing Web Agent Reasoning by Reconstructing Chain-of-Thought in Reflection, Branching, and Rollback. *arXiv preprint arXiv:2505.20013*, 2025. URL <https://arxiv.org/abs/2505.20013>.
 - [92] Muye Huang, Lingling Zhang, Jie Ma, Han Lai, Fangzhi Xu, Yifei Li, Wenjun Wu, Yaqiang Wu, and Jun Liu. ChartSketcher: Reasoning with Multimodal Feedback and Reflection for Chart Understanding. *arXiv preprint arXiv:2505.19076*, 2025. URL <https://arxiv.org/abs/2505.19076>.
 - [93] Sukai Huang, Trevor Cohn, and Nir Lipovetzky. Chasing Progress, Not Perfection: Revisiting Strategies for End-to-End LLM Plan Generation. *arXiv preprint arXiv:2412.10675*, 2024. URL <https://arxiv.org/abs/2412.10675>.
 - [94] Wenyang Hui and Kewei Tu. RoT: Enhancing Large Language Models with Reflection on Search Trees. *arXiv preprint arXiv:2404.05449*, 2024. URL <https://arxiv.org/abs/2404.05449>.
 - [95] Minbyul Jeong, Jiwoong Sohn, Mujeen Sung, and Jaewoo Kang. Improving Medical Reasoning through Retrieval and Self-Reflection with Retrieval-Augmented Large Language Models. *arXiv preprint arXiv:2401.15269*, 2024. URL <https://arxiv.org/abs/2401.15269>.
 - [96] Liya Ji, Chenyang Qi, and Qifeng Chen. Instruction-based Image Editing with Planning, Reasoning, and Generation. *arXiv preprint arXiv:2602.22624*, 2026. URL <https://arxiv.org/abs/2602.22624>.
 - [97] Shihao Ji, Zihui Song, Fucheng Zhong, Jisen Jia, Zhaobo Wu, Zheyi Cao, and Tianhao Xu. MyGO Multiplex CoT: A Method for Self-Reflection in Large Language Models via Double Chain of Thought Thinking. *arXiv preprint arXiv:2501.13117*, 2025. URL <https://arxiv.org/abs/2501.13117>.
 - [98] Xu Ji, Jianyi Zhang, Ziyin Zhou, Zhangchi Zhao, Qianqian Qiao, Kaiying Han, Md Imran Hossen, and Xiali Hei. Can't say cant? Measuring and Reasoning of Dark Jargons in Large Language Models. *arXiv preprint arXiv:2405.00718*, 2024. URL <https://arxiv.org/abs/2405.00718>.
 - [99] Ziwei Ji, Tiezheng Yu, Yan Xu, Nayeon Lee, Et-suko Ishii, and Pascale Fung. Towards Mitigating Hallucination in Large Language Models via Self-Reflection. *arXiv preprint arXiv:2310.06271*, 2023. URL <https://arxiv.org/abs/2310.06271>.
 - [100] Shuyi Jia, Chao Zhang, and Victor Fung. LL-MatDesign: Autonomous Materials Discovery with Large Language Models. *arXiv preprint arXiv:2406.13163*, 2024. URL <https://arxiv.org/abs/2406.13163>.
 - [101] Juyong Jiang, Jiasi Shen, Sunghun Kim, Kang Min Yoo, Jeonghoon Kim, and Sungju Kim. ReflexiCoder: Teaching Large Language Models to Self-Reflect on Generated Code and Self-Correct It via Reinforcement Learning. *arXiv preprint arXiv:2603.05863*, 2026. URL <https://arxiv.org/abs/2603.05863>.
 - [102] Xue Jiang, Yihong Dong, Lecheng Wang, Zheng Fang, Qiwei Shang, Ge Li, Zhi Jin, and Wenpin Jiao. Self-planning Code Generation with Large Language Models. *arXiv preprint arXiv:2303.06689*, 2023. URL <https://arxiv.org/abs/2303.06689>.
 - [103] Zhengbo Jiao, Shaobo Wang, Zifan Zhang, Xuan Ren, Wei Wang, Bing Zhao, Hu Wei, and Linfeng Zhang. Agentic Proposing: Enhancing Large Language Model Reasoning via Compositional Skill Synthesis. *arXiv preprint arXiv:2602.03279*, 2026. URL <https://arxiv.org/abs/2602.03279>.
 - [104] Frank Joubin, Antonello Ceravola, Pavel Smirnov, Felix Ocker, Joerg Deigmoeller, Anna Belardinelli, Chao Wang, Stephan Hasler, Daniel Tanneberg, and Michael Gienger. CoPAL: Corrective Planning of Robot Actions with Large Language Models. *arXiv preprint arXiv:2310.07263*, 2023. URL <https://arxiv.org/abs/2310.07263>.
 - [105] Liwei Kang, Yue Deng, Yao Xiao, Zhanfeng Mo, Wee Sun Lee, and Lidong Bing. First Try Matters: Revisiting the Role of Reflection in Reasoning Models. *arXiv preprint arXiv:2510.08308*, 2025. URL <https://arxiv.org/abs/2510.08308>.
 - [106] Shyam Sundar Kannan, Vishnunandan L. N. Venkatesh, and Byung-Cheol Min. SMART-LLM: Smart Multi-Agent Robot Task Planning using Large Language Models. *arXiv preprint arXiv:2309.10062*, 2023. URL <https://arxiv.org/abs/2309.10062>.
 - [107] Saketh Reddy Karra and Theja Tulabandhula. InteratSSort: Interactive Assortment Planning Us-

- ing Large Language Models. *arXiv preprint arXiv:2311.12241*, 2023. URL <https://arxiv.org/abs/2311.12241>.
- [108] Mohammad Khodadad, Ali Shiraei Kasmaee, Mahdi Astaraki, Nicholas Sherck, Hamidreza Mahyar, and Soheila Samiee. Evaluating Multi-Hop Reasoning in Large Language Models: A Chemistry-Centric Case Study. *arXiv preprint arXiv:2504.16414*, 2025. URL <https://arxiv.org/abs/2504.16414>.
- [109] Byeongjin Kim, Gyuwan Kim, and Seo Yeon Park. PPA-Plan: Proactive Pitfall Avoidance for Reliable Planning in Long-Context LLM Reasoning. *arXiv preprint arXiv:2601.11908*, 2026. URL <https://arxiv.org/abs/2601.11908>.
- [110] Yoonwoo Kim, Raghav Arora, Roberto Martin-Martin, Peter Stone, Ben Abbatemateo, and Yoonchang Sung. Large-Language-Model-Guided State Estimation for Partially Observable Task and Motion Planning. *arXiv preprint arXiv:2603.03704*, 2026. URL <https://arxiv.org/abs/2603.03704>.
- [111] Harsha Kokel, Michael Katz, Kavitha Srinivas, and Shirin Sohrabi. ACPBench: Reasoning about Action, Change, and Planning. *arXiv preprint arXiv:2410.05669*, 2024. URL <https://arxiv.org/abs/2410.05669>.
- [112] Xiangrui Kong, Wenxiao Zhang, Jin Hong, and Thomas Braunl. Embodied AI in Mobile Robots: Coverage Path Planning with Large Language Models. *arXiv preprint arXiv:2407.02220*, 2024. URL <https://arxiv.org/abs/2407.02220>.
- [113] Harsh Kumar, Ruiwei Xiao, Benjamin Lawson, Ilya Musabirov, Jiakai Shi, Xinyuan Wang, Huayin Luo, Joseph Jay Williams, Anna Rafferty, John Stamper, and Michael Liut. Supporting Self-Reflection at Scale with Large Language Models: Insights from Randomized Field Experiments in Classrooms. *arXiv preprint arXiv:2406.07571*, 2024. URL <https://arxiv.org/abs/2406.07571>.
- [114] Harsh Kumar, Suhyeon Yoo, Angela Zavaleta Bernuy, Jiakai Shi, Huayin Luo, Joseph Williams, Anastasia Kuzminykh, Ashton Anderson, and Rachel Kornfield. Large Language Model Agents for Improving Engagement with Behavior Change Interventions: Application to Digital Mindfulness. *arXiv preprint arXiv:2407.13067*, 2024. URL <https://arxiv.org/abs/2407.13067>.
- [115] Ce Li, Xiaofan Liu, Zhiyan Song, Ce Chi, Chen Zhao, Jingjing Yang, Zhendong Wang, Kexin Yang, Boshen Shi, Xing Wang, Chao Deng, and Junlan Feng. TReB: A Comprehensive Benchmark for Evaluating Table Reasoning Capabilities of Large Language Models. *arXiv preprint arXiv:2506.18421*, 2025. URL <https://arxiv.org/abs/2506.18421>.
- [116] Jia Li, Ge Li, Lecheng Wang, Hao Zhu, and Zhi Jin. Generating Equivalent Representations of Code By A Self-Reflection Approach. *arXiv preprint arXiv:2410.03351*, 2024. URL <https://arxiv.org/abs/2410.03351>.
- [117] Kaiyu Li, Zepeng Xin, Li Pang, Chao Pang, Yupeng Deng, Jing Yao, Guisong Xia, Deyu Meng, Zhi Wang, and Xiangyong Cao. SegEarth-R1: Geospatial Pixel Reasoning via Large Language Model. *arXiv preprint arXiv:2504.09644*, 2025. URL <https://arxiv.org/abs/2504.09644>.
- [118] Xiang Li, Ning Yan, and Masood Mortazavi. Embodied Task Planning via Graph-Informed Action Generation with Large Language Models. *arXiv preprint arXiv:2601.21841*, 2026. URL <https://arxiv.org/abs/2601.21841>.
- [119] Yanhong Li, Chenghao Yang, and Allyson Ettinger. When Hindsight is Not 20/20: Testing Limits on Reflective Thinking in Large Language Models. *arXiv preprint arXiv:2404.09129*, 2024. URL <https://arxiv.org/abs/2404.09129>.
- [120] Yu Li, Tian Lan, and Zhengling Qi. In-SPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization. *arXiv preprint arXiv:2512.23126*, 2025. URL <https://arxiv.org/abs/2512.23126>.
- [121] Feng Liang, Weixin Zeng, Runhao Zhao, and Xiang Zhao. NeSTR: A Neuro-Symbolic Abductive Framework for Temporal Reasoning in Large Language Models. *arXiv preprint arXiv:2512.07218*, 2025. URL <https://arxiv.org/abs/2512.07218>.
- [122] Jing Liang, Hongyao Tang, Yi Ma, Jinyi Liu, Yan Zheng, Shuyue Hu, Lei Bai, and Jianye Hao. Squeeze the Soaked Sponge: Efficient Off-policy Reinforcement Finetuning for Large Language Model. *arXiv preprint arXiv:2507.06892*, 2025. URL <https://arxiv.org/abs/2507.06892>.
- [123] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate. *arXiv preprint arXiv:2305.19118*, 2023. URL <https://arxiv.org/abs/2305.19118>.
- [124] Fangru Lin, Emanuele La Malfa, Valentin Hofmann, Elle Michelle Yang, Anthony Cohn, and Janet B. Pierrehumbert. Graph-enhanced Large Language Models in Asynchronous Plan Reasoning. *arXiv preprint arXiv:2402.02805*, 2024. URL <https://arxiv.org/abs/2402.02805>.
- [125] Hsien-Chin Lin, Benjamin Matthias Ruppik, Carel van Niekerk, Chia-Hao Shen, Michael Heck, Nurul Lubis, Renato Vukovic, Shutong Feng, and

- Milica Gasic. Prompt reinforcing for long-term planning of large language models. *arXiv preprint arXiv:2510.05921*, 2025. URL <https://arxiv.org/abs/2510.05921>.
- [126] Lixing Lin, Juli You, Yue Li, Luyun Lin, Yiqing Wang, Zhen Zhang, and Moxuan Zheng. Reflect-Guard: Enhancing LLM Safeguards against Adversarial Prompts via Logical Self-Reflection. *arXiv preprint arXiv:2605.24834*, 2026. URL <https://arxiv.org/abs/2605.24834>.
- [127] Wenjie Lin, Jin Wei-Kocsis, Jiansong Zhang, Byung-Cheol Min, Dongming Gan, Paul Asunda, and Ragu Athinarayanan. REFLEX: Metacognitive Reasoning for Reflective Zero-Shot Robotic Planning with Large Language Models. *arXiv preprint arXiv:2505.14899*, 2025. URL <https://arxiv.org/abs/2505.14899>.
- [128] Anthony Z. Liu, Xinhe Wang, Jacob Sansom, Yao Fu, Jongwook Choi, Sungryull Sohn, Jaekyeom Kim, and Honglak Lee. Interactive and Expressive Code-Augmented Planning with Large Language Models. *arXiv preprint arXiv:2411.13826*, 2024. URL <https://arxiv.org/abs/2411.13826>.
- [129] Chang Liu, Ying Chang, Jianmin Li, Yiqian Qu, Yu Li, Lingyong Cao, and Shuyuan Lin. Improving TCM Question Answering through Tree-Organized Self-Reflective Retrieval with LLMs. *arXiv preprint arXiv:2502.09156*, 2025. URL <https://arxiv.org/abs/2502.09156>.
- [130] Fengyuan Liu, Nouar AlDahoul, Gregory Eady, Yasir Zaki, and Talal Rahwan. Self-Reflection Makes Large Language Models Safer, Less Biased, and Ideologically Neutral. *arXiv preprint arXiv:2406.10400*, 2024. URL <https://arxiv.org/abs/2406.10400>.
- [131] Gang Liu, Michael Sun, Wojciech Matusik, Meng Jiang, and Jie Chen. Multimodal Large Language Models for Inverse Molecular Design with Retrosynthetic Planning. *arXiv preprint arXiv:2410.04223*, 2024. URL <https://arxiv.org/abs/2410.04223>.
- [132] Liping Liu, Chunhong Zhang, Likang Wu, Chuang Zhao, Zheng Hu, Ming He, and Jianping Fan. Instruct-of-Reflection: Enhancing Large Language Models Iterative Reflection Capabilities via Dynamic-Meta Instruction. *arXiv preprint arXiv:2503.00902*, 2025. URL <https://arxiv.org/abs/2503.00902>.
- [133] Qiang Liu, Xinlong Chen, Yue Ding, Bowen Song, Weiqiang Wang, Shu Wu, and Liang Wang. Attention-guided Self-reflection for Zero-shot Hallucination Detection in Large Language Models. *arXiv preprint arXiv:2501.09997*, 2025. URL <https://arxiv.org/abs/2501.09997>.
- [134] Wanlong Liu, Bo Zhang, Chenliang Li, Shaopeng Lai, Yuning Wu, Xuanyu Lei, and Ming Yan. R2-Write: Reflection and Revision for Open-Ended Writing with Deep Reasoning. *arXiv preprint arXiv:2604.03004*, 2026. URL <https://arxiv.org/abs/2604.03004>.
- [135] Wenru Liu, Pei Liu, and Jun Ma. DSDrive: Distilling Large Language Model for Lightweight End-to-End Autonomous Driving with Unified Reasoning and Planning. *arXiv preprint arXiv:2505.05360*, 2025. URL <https://arxiv.org/abs/2505.05360>.
- [136] Yaohua Liu. A Prompt-driven Task Planning Method for Multi-drones based on Large Language Model. *arXiv preprint arXiv:2406.00006*, 2024. URL <https://arxiv.org/abs/2406.00006>.
- [137] Yuchen Liu, Luigi Palmieri, Sebastian Koch, Ilche Georgievski, and Marco Aiello. DELTA: Decomposed Efficient Long-Term Robot Task Planning using Large Language Models. *arXiv preprint arXiv:2404.03275*, 2024. URL <https://arxiv.org/abs/2404.03275>.
- [138] Yuchen Liu, Luigi Palmieri, Sebastian Koch, Ilche Georgievski, and Marco Aiello. Towards Human Awareness in Robot Task Planning with Large Language Models. *arXiv preprint arXiv:2404.11267*, 2024. URL <https://arxiv.org/abs/2404.11267>.
- [139] Yuhang Liu, Pengxiang Li, Zishu Wei, Congkai Xie, Xueyu Hu, Xinchun Xu, Shengyu Zhang, Xiaotian Han, Hongxia Yang, and Fei Wu. InfiGU-IAgent: A Multimodal Generalist GUI Agent with Native Reasoning and Reflection. *arXiv preprint arXiv:2501.04575*, 2025. URL <https://arxiv.org/abs/2501.04575>.
- [140] Haolang Lu, Yilian Liu, Jingxin Xu, Guoshun Nan, Yuanlong Yu, Zhican Chen, and Kun Wang. Auditing Meta-Cognitive Hallucinations in Reasoning Large Language Models. *arXiv preprint arXiv:2505.13143*, 2025. URL <https://arxiv.org/abs/2505.13143>.
- [141] Hong Lu, Pierrick Lorang, Timothy R. Dugan, Jivko Sinapov, and Matthias Scheutz. Novelty Adaptation Through Hybrid Large Language Model (LLM)-Symbolic Planning and LLM-guided Reinforcement Learning. *arXiv preprint arXiv:2603.11351*, 2026. URL <https://arxiv.org/abs/2603.11351>.
- [142] Zhengdong Lu, Weikai Lu, Yiling Tao, Yun Dai, Zixuan Chen, Huiping Zhuang, Cen Chen, Hao Peng, and Ziqian Zeng. Decompose, Plan in Parallel, and Merge: A Novel Paradigm for Large Language Models based Planning with Multiple Constraints. *arXiv preprint arXiv:2506.02683*, 2025. URL <https://arxiv.org/abs/2506.02683>.
- [143] Kangyang Luo, Zichen Ding, Zhenmin Weng, Lingfeng Qiao, Meng Zhao, Xiang Li, Di Yin, and

- Jinlong Shu. Let's Be Self-generated via Step by Step: A Curriculum Learning Approach to Automated Reasoning with Large Language Models. *arXiv preprint arXiv:2410.21728*, 2024. URL <https://arxiv.org/abs/2410.21728>.
- [144] Linhao Luo, Yuan-Fang Li, Gholamreza Haffari, and Shirui Pan. Reasoning on Graphs: Faithful and Interpretable Large Language Model Reasoning. *arXiv preprint arXiv:2310.01061*, 2023. URL <https://arxiv.org/abs/2310.01061>.
- [145] Zhiyuan Ma, Jiayu Liu, Xianzhen Luo, Zhenya Huang, Qingfu Zhu, and Wanxiang Che. Advancing Tool-Augmented Large Language Models via Meta-Verification and Reflection Learning. *arXiv preprint arXiv:2506.04625*, 2025. URL <https://arxiv.org/abs/2506.04625>.
- [146] Tianyi Men, Pengfei Cao, Zhuoran Jin, Yubo Chen, Kang Liu, and Jun Zhao. Unlocking the Future: Exploring Look-Ahead Planning Mechanistic Interpretability in Large Language Models. *arXiv preprint arXiv:2406.16033*, 2024. URL <https://arxiv.org/abs/2406.16033>.
- [147] Jorge Mendez-Mendez. A Systematic Study of Large Language Models for Task and Motion Planning With PDDLStream. *arXiv preprint arXiv:2510.00182*, 2025. URL <https://arxiv.org/abs/2510.00182>.
- [148] Silin Meng, Yiwei Wang, Cheng-Fu Yang, Nanyun Peng, and Kai-Wei Chang. LLM-A*: Large Language Model Enhanced Incremental Heuristic Search on Path Planning. *arXiv preprint arXiv:2407.02511*, 2024. URL <https://arxiv.org/abs/2407.02511>.
- [149] Shentong Mo and Miao Xin. Tree of Uncertain Thoughts Reasoning for Large Language Models. *arXiv preprint arXiv:2309.07694*, 2023. URL <https://arxiv.org/abs/2309.07694>.
- [150] Dong Hyeon Mok, Seoin Back, Victor Fung, and Guoxiang Hu. Reasoning-Driven Design of Single Atom Catalysts via a Multi-Agent Large Language Model Framework. *arXiv preprint arXiv:2602.21533*, 2026. URL <https://arxiv.org/abs/2602.21533>.
- [151] Philipp Mondorf and Barbara Plank. Comparing Inferential Strategies of Humans and Large Language Models in Deductive Reasoning. *arXiv preprint arXiv:2402.14856*, 2024. URL <https://arxiv.org/abs/2402.14856>.
- [152] Sebastiano Monti, Carlo Nicolini, Gianni Pellegrini, Jacopo Staiano, and Bruno Lepri. SokoBench: Evaluating Long-Horizon Planning and Reasoning in Large Language Models. *arXiv preprint arXiv:2601.20856*, 2026. URL <https://arxiv.org/abs/2601.20856>.
- [153] Jared Moore, Rasmus Overmark, Ned Cooper, Beba Cibralic, Nick Haber, and Cameron R. Jones. Large Language Models Persuade Without Planning Theory of Mind. *arXiv preprint arXiv:2602.17045*, 2026. URL <https://arxiv.org/abs/2602.17045>.
- [154] Sajad Mousavi, Ricardo Luna Gutierrez, Desik Rengarajan, Vineet Gundecha, Ashwin Ramesh Babu, Avisek Naug, Antonio Guillen, and Soumyendu Sarkar. N-Critics: Self-Refinement of Large Language Models with Ensemble of Critics. *arXiv preprint arXiv:2310.18679*, 2023. URL <https://arxiv.org/abs/2310.18679>.
- [155] Jian Mu, Qixin Zhang, Zhiyong Wang, Menglin Yang, Shuang Qiu, Chengwei Qin, Zhongxiang Dai, and Yao Shu. Self-Reflective Generation at Test Time. *arXiv preprint arXiv:2510.02919*, 2025. URL <https://arxiv.org/abs/2510.02919>.
- [156] Muhammad Umair Nasir, Steven James, and Julian Togelius. GameTraversalBenchmark: Evaluating Planning Abilities Of Large Language Models Through Traversing 2D Game Maps. *arXiv preprint arXiv:2410.07765*, 2024. URL <https://arxiv.org/abs/2410.07765>.
- [157] Subigya Nepal, Arvind Pillai, William Campbell, Talie Massachi, Eunsol Soul Choi, Orson Xu, Joanna Kuc, Jeremy Huckins, Jason Holden, Colin Depp, Nicholas Jacobson, Mary Czerwinski, Eric Granholm, and Andrew T. Campbell. Contextual AI Journaling: Integrating LLM and Time Series Behavioral Sensing Technology to Promote Self-Reflection and Well-being using the MindScape App. *arXiv preprint arXiv:2404.00487*, 2024. URL <https://arxiv.org/abs/2404.00487>.
- [158] Chee Ng and Yuen Fung. Educational Personalized Learning Path Planning with Large Language Models. *arXiv preprint arXiv:2407.11773*, 2024. URL <https://arxiv.org/abs/2407.11773>.
- [159] Hang Ni, Fan Liu, Xinyu Ma, Lixin Su, Shuaiqiang Wang, Dawei Yin, Hui Xiong, and Hao Liu. TP-RAG: Benchmarking Retrieval-Augmented Large Language Model Agents for Spatiotemporal-Aware Travel Planning. *arXiv preprint arXiv:2504.08694*, 2025. URL <https://arxiv.org/abs/2504.08694>.
- [160] Jingwei Ni, Yu Fan, Vilem Zouhar, Donya Rooein, Alexander Hoyle, Mrinmaya Sachan, Markus Leopold, Dirk Hovy, and Elliott Ash. Can Reasoning Help Large Language Models Capture Human Annotator Disagreement? *arXiv preprint arXiv:2506.19467*, 2025. URL <https://arxiv.org/abs/2506.19467>.
- [161] Humza Nusrat, Luke Francisco, Bing Luo, Hassan Bagher-Ebadian, Joshua Kim, Karen Chinsnyder, Salim Siddiqui, Mira Shah, Eric Mellon, Mohammad Ghassemi, Anthony Doemer, Ben-

- jamin Movsas, and Kundan Thind. Automated stereotactic radiosurgery planning using a human-in-the-loop reasoning large language model agent. *arXiv preprint arXiv:2512.20586*, 2025. URL <https://arxiv.org/abs/2512.20586>.
- [162] James Oswald, Kavitha Srinivas, Harsha Kokel, Junkyu Lee, Michael Katz, and Shirin Sohrabi. Large Language Models as Planning Domain Generators. *arXiv preprint arXiv:2405.06650*, 2024. URL <https://arxiv.org/abs/2405.06650>.
- [163] Siqi Ouyang and Lei Li. AutoPlan: Automatic Planning of Interactive Decision-Making Tasks With Large Language Models. *arXiv preprint arXiv:2305.15064*, 2023. URL <https://arxiv.org/abs/2305.15064>.
- [164] Vishal Pallagani, Bharath Muppasani, Keerthiram Murugesan, Francesca Rossi, Biplav Srivastava, Lior Horesh, Francesco Fabiano, and Andrea Loreggia. Understanding the Capabilities of Large Language Models for Automated Planning. *arXiv preprint arXiv:2305.16151*, 2023. URL <https://arxiv.org/abs/2305.16151>.
- [165] Vishal Pallagani, Kaushik Roy, Bharath Muppasani, Francesco Fabiano, Andrea Loreggia, Keerthiram Murugesan, Biplav Srivastava, Francesca Rossi, Lior Horesh, and Amit Sheth. On the Prospects of Incorporating Large Language Models (LLMs) in Automated Planning and Scheduling (APS). *arXiv preprint arXiv:2401.02500*, 2024. URL <https://arxiv.org/abs/2401.02500>.
- [166] Aadi Palnitkar, Mingyang Mao, Nicholas Waytowich, Vinicius G. Goecks, and Xiaomin Lin. MILSCORE: Benchmarking Long-Context Geospatial Reasoning and Planning in Large Language Models. *arXiv preprint arXiv:2601.21826*, 2026. URL <https://arxiv.org/abs/2601.21826>.
- [167] Tanmay Parekh, Pradyot Prakash, Alexander Radovic, Akshay Shekher, and Denis Savenkov. Dynamic Strategy Planning for Efficient Question Answering with Large Language Models. *arXiv preprint arXiv:2410.23511*, 2024. URL <https://arxiv.org/abs/2410.23511>.
- [168] David Paulius, Alejandro Agostini, Benedict Quartey, and George Konidaris. Bootstrapping Object-level Planning with Large Language Models. *arXiv preprint arXiv:2409.12262*, 2024. URL <https://arxiv.org/abs/2409.12262>.
- [169] Hoang Phan, Victor Li, and Qi Lei. Think Twice, Generate Once: Safeguarding by Progressive Self-Reflection. *arXiv preprint arXiv:2510.01270*, 2025. URL <https://arxiv.org/abs/2510.01270>.
- [170] Shengmin Piao and Sanghyun Park. Tiny-Thinker: Distilling Reasoning through Coarse-to-Fine Knowledge Internalization with Self-Reflection. *arXiv preprint arXiv:2412.08024*, 2024. URL <https://arxiv.org/abs/2412.08024>.
- [171] Alexandre Piche, Aristides Milios, Dzmitry Bahdanau, and Chris Pal. LLMs can learn self-restraint through iterative self-reflection. *arXiv preprint arXiv:2405.13022*, 2024. URL <https://arxiv.org/abs/2405.13022>.
- [172] Aske Plaat, Annie Wong, Suzan Verberne, Joost Broekens, Niki van Stein, and Thomas Back. Multi-Step Reasoning with Large Language Models, a Survey. *arXiv preprint arXiv:2407.11511*, 2024. URL <https://arxiv.org/abs/2407.11511>.
- [173] Aske Plaat, Max van Duijn, Niki van Stein, Mike Preuss, Peter van der Putten, and Kees Joost Batenburg. Agentic Large Language Models, a survey. *arXiv preprint arXiv:2503.23037*, 2025. URL <https://arxiv.org/abs/2503.23037>.
- [174] Alistair Plum, Anne-Marie Lutgen, Christoph Purschke, and Achim Rettinger. Identity-Aware Large Language Models require Cultural Reasoning. *arXiv preprint arXiv:2510.18510*, 2025. URL <https://arxiv.org/abs/2510.18510>.
- [175] Nearchos Potamitis and Akhil Arora. Are Retrievals All You Need? Enhancing Large Language Model Reasoning Without Verbalized Feedback. *arXiv preprint arXiv:2504.12951*, 2025. URL <https://arxiv.org/abs/2504.12951>.
- [176] Parham Pourdavood, Michael Jacob, and Terrence Deacon. Large Language Models as symbolic DNA of cultural dynamics. *arXiv preprint arXiv:2506.21606*, 2025. URL <https://arxiv.org/abs/2506.21606>.
- [177] Israel Puerta-Merino, Carlos Nunez-Molina, Pablo Mesejo, and Juan Fernandez-Olivares. A Roadmap to Guide the Integration of LLMs in Hierarchical Planning. *arXiv preprint arXiv:2501.08068*, 2025. URL <https://arxiv.org/abs/2501.08068>.
- [178] Kai Qin, Liangxin Liu, Yu Liang, Longzheng Wang, Yan Wang, Yueyang Zhang, Long Xia, Zhiyuan Sun, Houde Liu, and Daiting Shi. ReflectRM: Boosting Generative Reward Models via Self-Reflection within a Unified Judgment Framework. *arXiv preprint arXiv:2604.07506*, 2026. URL <https://arxiv.org/abs/2604.07506>.
- [179] Lu Qiu, Yi Chen, Yuying Ge, Yixiao Ge, Ying Shan, and Xihui Liu. EgoPlan-Bench2: A Benchmark for Multimodal Large Language Model Planning in Real-World Scenarios. *arXiv preprint arXiv:2412.04447*, 2024. URL <https://arxiv.org/abs/2412.04447>.
- [180] Muhammad Reza Qorib, Qisheng Hu, and Hwee Tou Ng. Just What You Desire: Constrained Timeline Summarization with Self-Reflection for Enhanced Relevance. *arXiv preprint*

- arXiv:2412.17408*, 2024. URL <https://arxiv.org/abs/2412.17408>.
- [181] Hossein A. Rahmani, Satyapriya Krishna, Xi Wang, Mohammadmehdi Naghiaei, and Emine Yilmaz. Self-Correcting Large Language Models: Generation vs. Multiple Choice. *arXiv preprint arXiv:2511.09381*, 2025. URL <https://arxiv.org/abs/2511.09381>.
- [182] Allen Z. Ren, Brian Ichter, and Anirudha Majumdar. Thinking Forward and Backward: Effective Backward Planning with Large Language Models. *arXiv preprint arXiv:2411.01790*, 2024. URL <https://arxiv.org/abs/2411.01790>.
- [183] Matthew Renze and Erhan Guven. Self-Reflection in LLM Agents: Effects on Problem-Solving Performance. *arXiv preprint arXiv:2405.06682*, 2024. URL <https://arxiv.org/abs/2405.06682>.
- [184] Juan David Salazar Rodriguez, Sam Conrad Joyce, Nachamma Sockalingam, Khoo Eng Tat, and Julfendi. Student Perceptions of Large Language Models Use in Self-Reflection and Design Critique in Architecture Studio. *arXiv preprint arXiv:2602.00041*, 2026. URL <https://arxiv.org/abs/2602.00041>.
- [185] Rikard Rosenbacke, Carl Rosenbacke, Victor Rosenbacke, and Martin McKee. From Consumption to Reflection: Designing Human-AI Relations for Stable Reasoning. *arXiv preprint arXiv:2606.11195*, 2026. URL <https://arxiv.org/abs/2606.11195>.
- [186] Rikard Rosenbacke, Carl Rosenbacke, Victor Rosenbacke, and Martin McKee. Governing Reflective Human-AI Collaboration: A Framework for Epistemic Scaffolding and Traceable Reasoning. *arXiv preprint arXiv:2604.14898*, 2026. URL <https://arxiv.org/abs/2604.14898>.
- [187] Jessica Ryan, Alexander I. Gumilang, Robert William, and Derwin Suhartono. Self-MedRAG: a Self-Reflective Hybrid Retrieval-Augmented Generation Framework for Reliable Medical Question Answering. *arXiv preprint arXiv:2601.04531*, 2026. URL <https://arxiv.org/abs/2601.04531>.
- [188] Md Sadman Sakib and Yu Sun. Consolidating Trees of Robotic Plans Generated Using Large Language Models to Improve Reliability. *arXiv preprint arXiv:2401.07868*, 2024. URL <https://arxiv.org/abs/2401.07868>.
- [189] Charles Schepanowski and Charles Ling. On the Limits of Innate Planning in Large Language Models. *arXiv preprint arXiv:2511.21591*, 2025. URL <https://arxiv.org/abs/2511.21591>.
- [190] Martin Schroder. AutoScrum: Automating Project Planning Using Large Language Models. *arXiv preprint arXiv:2306.03197*, 2023. URL <https://arxiv.org/abs/2306.03197>.
- [191] Hongyu Shan, Mingyang Song, Chang Dai, Di Liang, and Han Chen. R-Capsule: Compressing High-Level Plans for Efficient Large Language Model Reasoning. *arXiv preprint arXiv:2509.22131*, 2025. URL <https://arxiv.org/abs/2509.22131>.
- [192] Vinay Sharma and Manish Jain. Enhancing reasoning accuracy in large language models during inference time. *arXiv preprint arXiv:2603.21301*, 2026. URL <https://arxiv.org/abs/2603.21301>.
- [193] Zijing Shi, Meng Fang, and Ling Chen. Monte Carlo Planning with Large Language Model for Text-Based Game Agents. *arXiv preprint arXiv:2504.16855*, 2025. URL <https://arxiv.org/abs/2504.16855>.
- [194] Dylan Shim and Minghan Wei. Constraint-aware Path Planning from Natural Language Instructions Using Large Language Models. *arXiv preprint arXiv:2603.19257*, 2026. URL <https://arxiv.org/abs/2603.19257>.
- [195] Donghoon Shin, Gary Hsieh, and Young-Ho Kim. PlanFitting: Personalized Exercise Planning with Large Language Model-driven Conversational Agent. *arXiv preprint arXiv:2309.12555*, 2023. URL <https://arxiv.org/abs/2309.12555>.
- [196] Safal Shrestha, Minwu Kim, and Keith Ross. Mathematical Reasoning in Large Language Models: Assessing Logical and Arithmetic Errors across Wide Numerical Ranges. *arXiv preprint arXiv:2502.08680*, 2025. URL <https://arxiv.org/abs/2502.08680>.
- [197] Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. ProgPrompt: Generating Situated Robot Task Plans using Large Language Models. *arXiv preprint arXiv:2209.11302*, 2022. URL <https://arxiv.org/abs/2209.11302>.
- [198] Chan Hee Song, Jiaman Wu, Clayton Washington, Brian M. Sadler, Wei-Lun Chao, and Yu Su. LLM-Planner: Few-Shot Grounded Planning for Embodied Agents with Large Language Models. *arXiv preprint arXiv:2212.04088*, 2022. URL <https://arxiv.org/abs/2212.04088>.
- [199] Jialin Song, Jonathan Raiman, and Bryan Catanzaro. Effective Large Language Model Debugging with Best-first Tree Search. *arXiv preprint arXiv:2407.19055*, 2024. URL <https://arxiv.org/abs/2407.19055>.
- [200] Yufan Song, Jiatao Zhang, Zeng Gu, Qingmiao Liang, Tuocheng Hu, Wei Song, and Shiqiang

- Zhu. FCRF: Flexible Constructivism Reflection for Long-Horizon Robotic Task Planning with Large Language Models. *arXiv preprint arXiv:2507.14975*, 2025. URL <https://arxiv.org/abs/2507.14975>.
- [201] Aseem Srivastava, Smriti Joshi, Tanmoy Chakraborty, and Md Shad Akhtar. Knowledge Planning in Large Language Models for Domain-Aligned Counseling Summarization. *arXiv preprint arXiv:2409.14907*, 2024. URL <https://arxiv.org/abs/2409.14907>.
- [202] Kaya Stechly, Karthik Valmeekam, and Subbarao Kambhampati. On the Self-Verification Limitations of Large Language Models on Reasoning and Planning Tasks. *arXiv preprint arXiv:2402.08115*, 2024. URL <https://arxiv.org/abs/2402.08115>.
- [203] Chris Su, Harrison Li, Matheus Marques, George Flint, Kevin Zhu, and Sunishchal Dev. Limits of Emergent Reasoning of Large Language Models in Agentic Frameworks for Deterministic Games. *arXiv preprint arXiv:2510.15974*, 2025. URL <https://arxiv.org/abs/2510.15974>.
- [204] Hao Sun, Hengyi Cai, Bo Wang, Yingyan Hou, Xiaochi Wei, Shuaiqiang Wang, Yan Zhang, and Dawei Yin. Towards Verifiable Text Generation with Evolving Memory and Self-Reflection. *arXiv preprint arXiv:2312.09075*, 2023. URL <https://arxiv.org/abs/2312.09075>.
- [205] Xiaoyu Tan, Haoyu Wang, Xihe Qiu, Yuan Cheng, Yinghui Xu, Wei Chu, and Yuan Qi. Struct-X: Enhancing Large Language Models Reasoning with Structured Data. *arXiv preprint arXiv:2407.12522*, 2024. URL <https://arxiv.org/abs/2407.12522>.
- [206] Yuzhe Tang. BrainBench: Exposing the Commonsense Reasoning Gap in Large Language Models. *arXiv preprint arXiv:2603.14761*, 2026. URL <https://arxiv.org/abs/2603.14761>.
- [207] Zhentao Tang, Yuqi Cui, Shixiong Kai, Wenqian Zhao, Ke Ye, Xing Li, Anxin Tian, Zehua Pei, Hui-Ling Zhen, Shoubo Hu, Xiaoguang Li, Yunhe Wang, and Mingxuan Yuan. ReThinker: Scientific Reasoning by Rethinking with Guided Reflection and Confidence Control. *arXiv preprint arXiv:2602.04496*, 2026. URL <https://arxiv.org/abs/2602.04496>.
- [208] Sree Harsha Tanneru, Dan Ley, Chirag Agarwal, and Himabindu Lakkaraju. On the Hardness of Faithful Chain-of-Thought Reasoning in Large Language Models. *arXiv preprint arXiv:2406.10625*, 2024. URL <https://arxiv.org/abs/2406.10625>.
- [209] Marcus Tantakoun, Xiaodan Zhu, and Christian Muise. LLMs as Planning Formalizers: A Survey for Leveraging Large Language Models to Construct Automated Planning Models. *arXiv preprint arXiv:2503.18971*, 2025. URL <https://arxiv.org/abs/2503.18971>.
- [210] Feng Tian, Flora D. Salim, and Hao Xue. TradingGroup: A Multi-Agent Trading System with Self-Reflection and Data-Synthesis. *arXiv preprint arXiv:2508.17565*, 2025. URL <https://arxiv.org/abs/2508.17565>.
- [211] Yao-Hung Hubert Tsai, Walter Talbott, and Jian Zhang. Efficient Non-Parametric Uncertainty Quantification for Black-Box Large Language Models and Decision Planning. *arXiv preprint arXiv:2402.00251*, 2024. URL <https://arxiv.org/abs/2402.00251>.
- [212] Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. PlanBench: An Extensible Benchmark for Evaluating Large Language Models on Planning and Reasoning about Change. *arXiv preprint arXiv:2206.10498*, 2022. URL <https://arxiv.org/abs/2206.10498>.
- [213] Karthik Valmeekam, Matthew Marquez, and Subbarao Kambhampati. Can Large Language Models Really Improve by Self-critiquing Their Own Plans? *arXiv preprint arXiv:2310.08118*, 2023. URL <https://arxiv.org/abs/2310.08118>.
- [214] Dirk Vath and Ngoc Thang Vu. Towards Zero-Shot, Controllable Dialog Planning with LLMs. *arXiv preprint arXiv:2410.05821*, 2024. URL <https://arxiv.org/abs/2410.05821>.
- [215] Matan Vetzler, Koren Lazar, Guy Uziel, Eran Hirsch, Ateret Anaby-Tavor, and Leshem Choshen. CRISP: Complex Reasoning with Interpretable Step-based Plans. *arXiv preprint arXiv:2507.08037*, 2025. URL <https://arxiv.org/abs/2507.08037>.
- [216] Xu Wan, Yansheng Wang, Wenqi Huang, and Mingyang Sun. Buffer Matters: Unleashing the Power of Off-Policy Reinforcement Learning in Large Language Model Reasoning. *arXiv preprint arXiv:2602.20722*, 2026. URL <https://arxiv.org/abs/2602.20722>.
- [217] Zhongwei Wan, Zhihao Dou, Che Liu, Yu Zhang, Dongfei Cui, Qinqian Zhao, Hui Shen, Jing Xiong, Yi Xin, Yifan Jiang, Chaofan Tao, Yangfan He, Mi Zhang, and Shen Yan. SRPO: Enhancing Multimodal LLM Reasoning via Reflection-Aware Reinforcement Learning. *arXiv preprint arXiv:2506.01713*, 2025. URL <https://arxiv.org/abs/2506.01713>.
- [218] Danqing Wang, Zhuorui Ye, Fei Fang, and Lei Li. Cooperative Strategic Planning Enhances Reasoning Capabilities in Large Language Models. *arXiv preprint arXiv:2410.20007*, 2024. URL <https://arxiv.org/abs/2410.20007>.

- [219] Dingrui Wang, Marc Kaufeld, and Johannes Betz. DualAD: Dual-Layer Planning for Reasoning in Autonomous Driving. *arXiv preprint arXiv:2409.18053*, 2024. URL <https://arxiv.org/abs/2409.18053>.
- [220] Hanyu Wang, Yuanpu Cao, Lu Lin, and Jinghui Chen. PreFlect: From Retrospective to Prospective Reflection in Large Language Model Agents. *arXiv preprint arXiv:2602.07187*, 2026. URL <https://arxiv.org/abs/2602.07187>.
- [221] Jiahao Wang, Songkai Xue, Jinghui Li, and Xiaozhen Wang. Diverse Human Value Alignment for Large Language Models via Ethical Reasoning. *arXiv preprint arXiv:2511.00379*, 2025. URL <https://arxiv.org/abs/2511.00379>.
- [222] Jingyuan Wang, Yankai Chen, Zhonghang Li, and Chao Huang. LightReasoner: Can Small Language Models Teach Large Language Models Reasoning? *arXiv preprint arXiv:2510.07962*, 2025. URL <https://arxiv.org/abs/2510.07962>.
- [223] Junjie Wang, Mingyang Chen, Binbin Hu, Dan Yang, Ziqi Liu, Yue Shen, Peng Wei, Zhiqiang Zhang, Jinjie Gu, Jun Zhou, Jeff Z. Pan, Wen Zhang, and Huajun Chen. Learning to Plan for Retrieval-Augmented Large Language Models from Knowledge Graphs. *arXiv preprint arXiv:2406.14282*, 2024. URL <https://arxiv.org/abs/2406.14282>.
- [224] Kevin Wang, Junbo Li, Neel P. Bhatt, Yihan Xi, Qiang Liu, Ufuk Topcu, and Zhangyang Wang. On The Planning Abilities of OpenAI’s o1 Models: Feasibility, Optimality, and Generalizability. *arXiv preprint arXiv:2409.19924*, 2024. URL <https://arxiv.org/abs/2409.19924>.
- [225] Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. Plan-and-Solve Prompting: Improving Zero-Shot Chain-of-Thought Reasoning by Large Language Models. *arXiv preprint arXiv:2305.04091*, 2023. URL <https://arxiv.org/abs/2305.04091>.
- [226] Liman Wang and Hanyang Zhong. LLM-SAP: Large Language Models Situational Awareness Based Planning. *arXiv preprint arXiv:2312.16127*, 2023. URL <https://arxiv.org/abs/2312.16127>.
- [227] Shu Wang, Muzhi Han, Ziyuan Jiao, Zeyu Zhang, Ying Nian Wu, Song-Chun Zhu, and Hangxin Liu. LLM3: Large Language Model-based Task and Motion Planning with Motion Failure Reasoning. *arXiv preprint arXiv:2403.11552*, 2024. URL <https://arxiv.org/abs/2403.11552>.
- [228] Teng Wang, Zhangyi Jiang, Zhenqi He, Shenyang Tong, Wenhan Yang, Yanan Zheng, Zeyu Li, Zifan He, Hailei Gong, Zewen Ye, Shengjie Ma, and Jianping Zhang. Towards Hierarchical Multi-Step Reward Models for Enhanced Reasoning in Large Language Models. *arXiv preprint arXiv:2503.13551*, 2025. URL <https://arxiv.org/abs/2503.13551>.
- [229] Tianlong Wang, Pinqiao Wang, Weili Shi, and Sheng li. ItinBench: Benchmarking Planning Across Multiple Cognitive Dimensions with Large Language Models. *arXiv preprint arXiv:2603.19515*, 2026. URL <https://arxiv.org/abs/2603.19515>.
- [230] Xinyi Wang, Lucas Caccia, Oleksiy Ostapenko, Xingdi Yuan, William Yang Wang, and Alessandro Sordoni. Guiding Language Model Reasoning with Planning Tokens. *arXiv preprint arXiv:2310.05707*, 2023. URL <https://arxiv.org/abs/2310.05707>.
- [231] Xuan Wang, Yu Ming, Xinhao Zhong, Xinyu Yu, Wenjie Wang, Shuai Chen, and Wei Lin. SPREG: Structured Plan Repair with Entropy-Guided Test-Time Intervention for Large Language Model Reasoning. *arXiv preprint arXiv:2604.17884*, 2026. URL <https://arxiv.org/abs/2604.17884>.
- [232] Yiping Wang, Qing Yang, Zhiyuan Zeng, Liliang Ren, Liyuan Liu, Baolin Peng, Hao Cheng, Xuehai He, Kuan Wang, Jianfeng Gao, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. Reinforcement Learning for Reasoning in Large Language Models with One Training Example. *arXiv preprint arXiv:2504.20571*, 2025. URL <https://arxiv.org/abs/2504.20571>.
- [233] Yutong Wang, Jiali Zeng, Xuebo Liu, Fandong Meng, Jie Zhou, and Min Zhang. TasTe: Teaching Large Language Models to Translate through Self-Reflection. *arXiv preprint arXiv:2406.08434*, 2024. URL <https://arxiv.org/abs/2406.08434>.
- [234] Zehong Wang, Fang Wu, Hongru Wang, Xiangru Tang, Bolian Li, Zhenfei Yin, Yijun Ma, Yiyang Li, Weixiang Sun, Xiushi Chen, and Yanfang Ye. Why Reasoning Fails to Plan: A Planning-Centric Analysis of Long-Horizon Decision Making in LLM Agents. *arXiv preprint arXiv:2601.22311*, 2026. URL <https://arxiv.org/abs/2601.22311>.
- [235] Zhaoyang Wang, Shaohan Huang, Yuxuan Liu, Jiahai Wang, Minghui Song, Zihan Zhang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, and Qi Zhang. Democratizing Reasoning Ability: Tailored Learning from Large Language Model. *arXiv preprint arXiv:2310.13332*, 2023. URL <https://arxiv.org/abs/2310.13332>.
- [236] Zhijie Wang. GRPO and Reflection Reward for Mathematical Reasoning in Large Language Models. *arXiv preprint arXiv:2603.14041*, 2026. URL <https://arxiv.org/abs/2603.14041>.
- [237] Ziyan Wang, Yingpeng Du, Zhu Sun, Haoyan Chua, Kaidong Feng, Wenya Wang, and Jie Zhang. Re2LLM: Reflective Reinforcement Large Language Model for Session-based Recommendation.

- arXiv preprint arXiv:2403.16427*, 2024. URL <https://arxiv.org/abs/2403.16427>.
- [238] Sion Weatherhead, Flora Salim, and Aaron Belbasis. Illusions of reflection: open-ended task reveals systematic failures in Large Language Models’ reflective reasoning. *arXiv preprint arXiv:2510.18254*, 2025. URL <https://arxiv.org/abs/2510.18254>.
- [239] Taylor Webb, Shanka Subhra Mondal, and Ida Momennejad. Improving Planning with Large Language Models: A Modular Agentic Architecture. *arXiv preprint arXiv:2310.00194*, 2023. URL <https://arxiv.org/abs/2310.00194>.
- [240] Tianxin Wei, Ting-Wei Li, Zhining Liu, Xuying Ning, Ze Yang, Jiaru Zou, Zhichen Zeng, Ruizhong Qiu, Xiao Lin, Dongqi Fu, Zihao Li, Mengting Ai, Duo Zhou, Wenxuan Bao, Yunzhe Li, Gaotang Li, Cheng Qian, Yu Wang, Xiangru Tang, Yin Xiao, et al. Agentic Reasoning for Large Language Models. *arXiv preprint arXiv:2601.12538*, 2026. URL <https://arxiv.org/abs/2601.12538>.
- [241] Annie Wong, Thomas Back, Aske Plaat, Niki van Stein, and Anna V. Kononova. Reasoning Capabilities of Large Language Models on Dynamic Tasks. *arXiv preprint arXiv:2505.10543*, 2025. URL <https://arxiv.org/abs/2505.10543>.
- [242] Duo Wu, Jinghe Wang, Yuan Meng, Yanning Zhang, Le Sun, and Zhi Wang. CATP-LLM: Empowering Large Language Models for Cost-Aware Tool Planning. *arXiv preprint arXiv:2411.16313*, 2024. URL <https://arxiv.org/abs/2411.16313>.
- [243] Jimmy Wu, Rika Antonova, Adam Kan, Marion Lepert, Andy Zeng, Shuran Song, Jeanette Bohg, Szymon Rusinkiewicz, and Thomas Funkhouser. TidyBot: Personalized Robot Assistance with Large Language Models. *arXiv preprint arXiv:2305.05658*, 2023. URL <https://arxiv.org/abs/2305.05658>.
- [244] Penghao Wu, Shengnan Ma, Bo Wang, Jiaheng Yu, Lewei Lu, and Ziwei Liu. GUI-Reflection: Empowering Multimodal GUI Models with Self-Reflection Behavior. *arXiv preprint arXiv:2506.08012*, 2025. URL <https://arxiv.org/abs/2506.08012>.
- [245] Zhenyu Wu, Ziwei Wang, Xiuwei Xu, Jiwen Lu, and Haibin Yan. Embodied Task Planning with Large Language Models. *arXiv preprint arXiv:2307.01848*, 2023. URL <https://arxiv.org/abs/2307.01848>.
- [246] Zhuoyang Wu, Xinze Li, Zhenghao Liu, Yukun Yan, Zhiyuan Liu, Minghe Yu, Cheng Yang, Yu Gu, Ge Yu, and Maosong Sun. Enhancing Long-Chain Reasoning Distillation through Error-Aware Self-Reflection. *arXiv preprint arXiv:2505.22131*, 2025. URL <https://arxiv.org/abs/2505.22131>.
- [247] Wenke Xia, Ruoxuan Feng, Dong Wang, and Di Hu. Phoenix: A Motion-based Self-Reflection Framework for Fine-grained Robotic Action Correction. *arXiv preprint arXiv:2504.14588*, 2025. URL <https://arxiv.org/abs/2504.14588>.
- [248] Chengxing Xie and Difan Zou. A Human-Like Reasoning Framework for Multi-Phases Planning Task with Large Language Models. *arXiv preprint arXiv:2405.18208*, 2024. URL <https://arxiv.org/abs/2405.18208>.
- [249] Yaqi Xie, Chen Yu, Tongyao Zhu, Jinbin Bai, Ze Gong, and Harold Soh. Translating Natural Language to Planning Goals with Large-Language Models. *arXiv preprint arXiv:2302.05128*, 2023. URL <https://arxiv.org/abs/2302.05128>.
- [250] Amy Xin, Jinxin Liu, Zijun Yao, Zhicheng Lee, Shulin Cao, Lei Hou, and Juanzi Li. AtomR: Atomic Operator-Empowered Large Language Models for Heterogeneous Knowledge Reasoning. *arXiv preprint arXiv:2411.16495*, 2024. URL <https://arxiv.org/abs/2411.16495>.
- [251] Sishi Xiong, Dakai Wang, Yu Zhao, Jie Zhang, Changzai Pan, Haowei He, Xiangyu Li, Wenhan Chang, Zhongjiang He, Shuangyong Song, and Yongxiang Li. TableReasoner: Advancing Table Reasoning Framework with Large Language Models. *arXiv preprint arXiv:2507.08046*, 2025. URL <https://arxiv.org/abs/2507.08046>.
- [252] Fengli Xu, Qianyu Hao, Zefang Zong, Jingwei Wang, Yunke Zhang, Jingyi Wang, Xiaochong Lan, Jiahui Gong, Tianjian Ouyang, Fanjin Meng, Chenyang Shao, Yuwei Yan, Qinglong Yang, Yiwen Song, Sijian Ren, Xinyuan Hu, Yu Li, Jie Feng, Chen Gao, and Yong Li. Towards Large Reasoning Models: A Survey of Reinforced Reasoning with Large Language Models. *arXiv preprint arXiv:2501.09686*, 2025. URL <https://arxiv.org/abs/2501.09686>.
- [253] Hongshen Xu, Zixv yang, Zichen Zhu, Kunyao Lan, Zihan Wang, Mengyue Wu, Ziwei Ji, Lu Chen, Pascale Fung, and Kai Yu. Delusions of Large Language Models. *arXiv preprint arXiv:2503.06709*, 2025. URL <https://arxiv.org/abs/2503.06709>.
- [254] Jinxuan Xu, Shiyu Jin, Yutian Lei, Yuqian Zhang, and Liangjun Zhang. RT-Grasp: Reasoning Tuning Robotic Grasping via Multi-modal Large Language Model. *arXiv preprint arXiv:2411.05212*, 2024. URL <https://arxiv.org/abs/2411.05212>.
- [255] Mengya Xu, Zhongzhen Huang, Jie Zhang, Xiaofan Zhang, and Qi Dou. Surgical Action Planning with Large Language Models. *arXiv preprint arXiv:2503.18296*, 2025. URL <https://arxiv.org/abs/2503.18296>.
- [256] Yifei Xu, Tusher Chakraborty, Srinagesh Sharma, Leonardo Nunes, Swati Sharma, Kate Drakos De-

- mopolos, Emre Kcman, Songwu Lu, and Ranveer Chandra. Direct Reasoning Optimization: Token-Level Reasoning Reflectivity Meets Rubric Gates for Unverifiable Tasks. *arXiv preprint arXiv:2506.13351*, 2025. URL <https://arxiv.org/abs/2506.13351>.
- [257] Christos Xypolopoulos, Guokan Shang, Xiao Fei, Giannis Nikolentzos, Hadi Abdine, Iakovos Evdaimon, Michail Chatzianastasis, Giorgos Stamou, and Michalis Vazirgiannis. Graph Linearization Methods for Reasoning on Graphs with Large Language Models. *arXiv preprint arXiv:2410.19494*, 2024. URL <https://arxiv.org/abs/2410.19494>.
- [258] Haijiang Yan, Jian-Qiao Zhu, and Adam Sanborn. Large Language Models Can Take False First Steps at Inference-time Planning. *arXiv preprint arXiv:2602.02991*, 2026. URL <https://arxiv.org/abs/2602.02991>.
- [259] Hanqi Yan, Qinglin Zhu, Xinyu Wang, Lin Gui, and Yulan He. Mirror: A Multiple-perspective Self-Reflection Method for Knowledge-rich Reasoning. *arXiv preprint arXiv:2402.14963*, 2024. URL <https://arxiv.org/abs/2402.14963>.
- [260] Bo Yang, Jiaxian Guo, Yusuke Iwasawa, and Yutaka Matsuo. Large Language Models as Theory of Mind Aware Generative Agents with Counterfactual Reflection. *arXiv preprint arXiv:2501.15355*, 2025. URL <https://arxiv.org/abs/2501.15355>.
- [261] Dayu Yang, Tianyang Liu, Daoan Zhang, Antoine Simoulin, Xiaoyi Liu, Yuwei Cao, Zhaopu Teng, Xin Qian, Grey Yang, Jiebo Luo, and Julian McAuley. Code to Think, Think to Code: A Survey on Code-Enhanced Reasoning and Reasoning-Driven Code Intelligence in LLMs. *arXiv preprint arXiv:2502.19411*, 2025. URL <https://arxiv.org/abs/2502.19411>.
- [262] Dejie Yang, Zijing Zhao, and Yang Liu. Plan-LLM: Video Procedure Planning with Refinable Large Language Models. *arXiv preprint arXiv:2412.19139*, 2024. URL <https://arxiv.org/abs/2412.19139>.
- [263] Yunhao Yang and Anshul Tomar. On the Planning, Search, and Memorization Capabilities of Large Language Models. *arXiv preprint arXiv:2309.01868*, 2023. URL <https://arxiv.org/abs/2309.01868>.
- [264] Zhun Yang, Adam Ishay, and Joohyung Lee. Coupling Large Language Models with Logic Programming for Robust and General Reasoning from Text. *arXiv preprint arXiv:2307.07696*, 2023. URL <https://arxiv.org/abs/2307.07696>.
- [265] Shaowei Ying, Zhenlong Li, and Manzhu Yu. Beyond Words: Evaluating Large Language Models in Transportation Planning. *arXiv preprint arXiv:2409.14516*, 2024. URL <https://arxiv.org/abs/2409.14516>.
- [266] Yahan Yu, Noa Nakanishi, and Fei Cheng. Revisiting Anthropomorphic Reflection Markers in Large Language Model Reasoning. *arXiv preprint arXiv:2605.28305*, 2026. URL <https://arxiv.org/abs/2605.28305>.
- [267] Zhongwei Yu, Wannian Xia, Xue Yan, Bo Xu, Haifeng Zhang, Yali Du, and Jun Wang. Self-Verifying Reflection Helps Transformers with CoT Reasoning. *arXiv preprint arXiv:2510.12157*, 2025. URL <https://arxiv.org/abs/2510.12157>.
- [268] Siyu Yuan, Jiangjie Chen, Ziquan Fu, Xuyang Ge, Soham Shah, Charles Robert Jankowski, Yanghua Xiao, and Deqing Yang. Distilling Script Knowledge from Large Language Models for Constrained Language Planning. *arXiv preprint arXiv:2305.05252*, 2023. URL <https://arxiv.org/abs/2305.05252>.
- [269] Yuanhao Yue, Chengyu Wang, Jun Huang, and Peng Wang. Distilling Instruction-following Abilities of Large Language Models with Task-aware Curriculum Planning. *arXiv preprint arXiv:2405.13448*, 2024. URL <https://arxiv.org/abs/2405.13448>.
- [270] Qinglin Zeng, Jing Yang, and Keze Wang. Reflective Confidence: Correcting Reasoning Flaws via Online Self-Correction. *arXiv preprint arXiv:2512.18605*, 2025. URL <https://arxiv.org/abs/2512.18605>.
- [271] Zaifu Zhan, Mengyuan Cui, and Rui Zhang. Can Large Language Models Self-Correct in Medical Question Answering? An Exploratory Study. *arXiv preprint arXiv:2604.00261*, 2026. URL <https://arxiv.org/abs/2604.00261>.
- [272] Borui Zhang, Chaojie Li, Guo Chen, and Zhaoyang Dong. Large Language Model Assisted Optimal Bidding of BESS in FCAS Market: An AI-agent based Approach. *arXiv preprint arXiv:2406.00974*, 2024. URL <https://arxiv.org/abs/2406.00974>.
- [273] Ce Zhang, Jinxi He, Junyi He, Katia Sycara, and Yaqi Xie. Evolving Contextual Safety in Multi-Modal Large Language Models via Inference-Time Self-Reflective Memory. *arXiv preprint arXiv:2603.15800*, 2026. URL <https://arxiv.org/abs/2603.15800>.
- [274] Han Zhang, Langshi Zhou, and Hanfang Yang. Learning to Retrieve and Reason on Knowledge Graph through Active Self-Reflection. *arXiv preprint arXiv:2502.14932*, 2025. URL <https://arxiv.org/abs/2502.14932>.
- [275] Jingshen Zhang, Bo Wang, Boci Yang, Dongming Zhao, Ruifang He, Yuxian Hou, and Zifei Yu. Understanding the Self-Reflection Mechanisms of

- LLMs through Biased Attitude Associations. *arXiv preprint arXiv:2606.00600*, 2026. URL <https://arxiv.org/abs/2606.00600>.
- [276] Li Zhang. Structured Event Reasoning with Large Language Models. *arXiv preprint arXiv:2408.16098*, 2024. URL <https://arxiv.org/abs/2408.16098>.
- [277] Shun Zhang, Zhenfang Chen, Yikang Shen, Mingyu Ding, Joshua B. Tenenbaum, and Chuang Gan. Planning with Large Language Models for Code Generation. *arXiv preprint arXiv:2303.05510*, 2023. URL <https://arxiv.org/abs/2303.05510>.
- [278] Tao Zhang, Kaixian Qu, Zhibin Li, Jiajun Wu, Marco Hutter, Manling Li, and Fan Shi. Using large language models for embodied planning introduces systematic safety risks. *arXiv preprint arXiv:2604.18463*, 2026. URL <https://arxiv.org/abs/2604.18463>.
- [279] Taolin Zhang, Zihan Ma, Maosong Cao, Junnan Liu, Songyang Zhang, and Kai Chen. Coding Triangle: How Does Large Language Model Understand Code? *arXiv preprint arXiv:2507.06138*, 2025. URL <https://arxiv.org/abs/2507.06138>.
- [280] Xudong Zhang, Jian Yang, Shengkai Wang, Jiangpeng Tian, Shaowen Chen, Xian Wei, Ke Li, and Xiong You. The Sword, Shield, and Achilles’ Heel: Characterizing the Linguistic Inductive Bias of Large Language Models for Spatial Reasoning in Navigation Planning. *arXiv preprint arXiv:2605.31404*, 2026. URL <https://arxiv.org/abs/2605.31404>.
- [281] Zhenyu Zhang, Tianyi Chen, Weiran Xu, Alex Pentland, and Jiaxin Pei. ReCAP: Recursive Context-Aware Reasoning and Planning for Large Language Model Agents. *arXiv preprint arXiv:2510.23822*, 2025. URL <https://arxiv.org/abs/2510.23822>.
- [282] Chengshuai Zhao, Zhen Tan, Pingchuan Ma, Dawei Li, Bohan Jiang, Yancheng Wang, Yingzhen Yang, and Huan Liu. Is Chain-of-Thought Reasoning of LLMs a Mirage? A Data Distribution Lens. *arXiv preprint arXiv:2508.01191*, 2025. URL <https://arxiv.org/abs/2508.01191>.
- [283] Siyan Zhao, Devaansh Gupta, Qinqing Zheng, and Aditya Grover. d1: Scaling Reasoning in Diffusion Large Language Models via Reinforcement Learning. *arXiv preprint arXiv:2504.12216*, 2025. URL <https://arxiv.org/abs/2504.12216>.
- [284] Yachao Zhao, Bo Wang, Yan Wang, Dongming Zhao, Ruifang He, and Yuexian Hou. Explicit vs. Implicit: Investigating Social Bias in Large Language Models through Self-Reflection. *arXiv preprint arXiv:2501.02295*, 2025. URL <https://arxiv.org/abs/2501.02295>.
- [285] Yusheng Zhao, Xiao Luo, Weizhi Zhang, Wei Ju, Zhiping Xiao, Philip S. Yu, and Ming Zhang. MARCO: Meta-Reflection with Cross-Referencing for Code Reasoning. *arXiv preprint arXiv:2505.17481*, 2025. URL <https://arxiv.org/abs/2505.17481>.
- [286] Zibo Zhao, Yuanting Zha, Haipeng Zhang, and Xingcheng Xu. The Two-Stage Decision-Sampling Hypothesis: Understanding the Emergence of Self-Reflection in RL-Trained LLMs. *arXiv preprint arXiv:2601.01580*, 2026. URL <https://arxiv.org/abs/2601.01580>.
- [287] Zirui Zhao, Wee Sun Lee, and David Hsu. Large Language Models as Commonsense Knowledge for Large-Scale Task Planning. *arXiv preprint arXiv:2305.14078*, 2023. URL <https://arxiv.org/abs/2305.14078>.
- [288] Haoyu Zheng, Yun Zhu, Yuqian Yuan, Bo Yuan, Wenqiao Zhang, Siliang Tang, and Jun Xiao. PILOT: Planning via Internalized Latent Optimization Trajectories for Large Language Models. *arXiv preprint arXiv:2601.19917*, 2026. URL <https://arxiv.org/abs/2601.19917>.
- [289] Kaizhi Zheng, Kaiwen Zhou, Jing Gu, Yue Fan, Jialu Wang, Zonglin Di, Xuehai He, and Xin Eric Wang. JARVIS: A Neuro-Symbolic Commonsense Reasoning Framework for Conversational Embodied Agents. *arXiv preprint arXiv:2208.13266*, 2022. URL <https://arxiv.org/abs/2208.13266>.
- [290] Yu Zheng, Longyi Liu, Yuming Lin, Jie Feng, Guozhen Zhang, Depeng Jin, and Yong Li. UrbanPlanBench: A Comprehensive Urban Planning Benchmark for Evaluating Large Language Models. *arXiv preprint arXiv:2504.21027*, 2025. URL <https://arxiv.org/abs/2504.21027>.
- [291] Gengze Zhou, Yicong Hong, and Qi Wu. NavGPT: Explicit Reasoning in Vision-and-Language Navigation with Large Language Models. *arXiv preprint arXiv:2305.16986*, 2023. URL <https://arxiv.org/abs/2305.16986>.
- [292] Weijie Zhou, Manli Tao, Chaoyang Zhao, Honghui Dong, Ming Tang, and Jinqiao Wang. Light-Planner: Unleashing the Reasoning Capabilities of Lightweight Large Language Models in Task Planning. *arXiv preprint arXiv:2503.08508*, 2025. URL <https://arxiv.org/abs/2503.08508>.
- [293] Zhilun Zhou, Yuming Lin, and Yong Li. Large language model empowered participatory urban planning. *arXiv preprint arXiv:2402.01698*, 2024. URL <https://arxiv.org/abs/2402.01698>.
- [294] Jiajun Zhu, Ye Liu, Meikai Bao, Kai Zhang, Yang-hai Zhang, and Qi Liu. Self-Reflective Planning with Knowledge Graphs: Enhancing LLM Reasoning Reliability for Question Answering. *arXiv*

- preprint arXiv:2505.19410*, 2025. URL <https://arxiv.org/abs/2505.19410>.
- [295] Qin Zhu, Fei Huang, Runyu Peng, Keming Lu, Bowen Yu, Qinyuan Cheng, Xipeng Qiu, Xuanjing Huang, and Junyang Lin. AutoLogi: Automated Generation of Logic Puzzles for Evaluating Reasoning Abilities of Large Language Models. *arXiv preprint arXiv:2502.16906*, 2025. URL <https://arxiv.org/abs/2502.16906>.
 - [296] Wang Bill Zhu, Miaosen Chai, Ishika Singh, Robin Jia, and Jesse Thomason. PSALM-V: Automating Symbolic Planning in Interactive Visual Environments with Large Language Models. *arXiv preprint arXiv:2506.20097*, 2025. URL <https://arxiv.org/abs/2506.20097>.
 - [297] Xudong Zhu, Jiachen Jiang, Mohammad Mahdi Khalili, and Zhihui Zhu. From Emergence to Control: Probing and Modulating Self-Reflection in Language Models. *arXiv preprint arXiv:2506.12217*, 2025. URL <https://arxiv.org/abs/2506.12217>.
 - [298] Xuyang Zhu, Sejoon Chang, and Andrew Kuik. Enhancing Critical Thinking with AI: A Tailored Warning System for RAG Models. *arXiv preprint arXiv:2504.16883*, 2025. URL <https://arxiv.org/abs/2504.16883>.
 - [299] Yin Zhu, Zhiling Luo, and Gong Cheng. Furthest Reasoning with Plan Assessment: Stable Reasoning Path with Retrieval-Augmented Large Language Models. *arXiv preprint arXiv:2309.12767*, 2023. URL <https://arxiv.org/abs/2309.12767>.
 - [300] Haomin Zhuang, Hojun Yoo, Xiaonan Luo, Kehan Guo, and Xiangliang Zhang. Reliable Control-Point Selection for Steering Reasoning in Large Language Models. *arXiv preprint arXiv:2604.02113*, 2026. URL <https://arxiv.org/abs/2604.02113>.